

THE UNIVERSITY OF NEW SOUTH WALES
SCHOOL OF PHYSICS
THIRD YEAR LABORATORY

HOLOGRAPHY

$H(\quad)$

**PLEASE RETURN THIS
FOLDER TO THE LAB**

FOLDER CONTENTS

Holography

1. **Instructions for processing holographic plates**
2. **Kasper J.E. and Feller S.A., "The Complete Book of Holograms" (1986) - extracts from Chs 1, 4, 5, 6, 7 and 8**
3. **O'Shea D.C., Callen W.R. and Rhodes W.T., "Introduction to Lasers and their Applications" (1978) pp. 198 - 212**
4. **Hariharan P., "Simplified processing techniques for holography" Phot. Sci. and Eng. 24, 105 (1980)**

HOLOGRAM PROCESSING PROCEDURES

WARNINGS!

- Keep the **extractor fan on at all times** while working in the dark room.
- When using photographic chemicals, **lab coats** and **disposable gloves** should be worn to prevent contact between the chemicals and your skin, or clothes.

Setting up

Check the dates on the bottles of premixed **developer** and **fixer**. If either solution is greater than 6 months old, or if the bleach has run out, see your demonstrator.

Use the plastic 15x20cm trays for developer, fixer and bleach (trays are appropriately labelled). You will need about 250ml (~10mm depth) of solution in each tray. Tongs should be used to help manoeuvre the plates in the trays. Use a large tray in the sink for the washing steps.

Line the trays up in the order in which you will use them; set appropriate times on the timers and familiarise yourself with the location of all objects relevant to the processing before you start.

Before commencing the processing **lock the outside door** and close the inner door so that people can't accidentally enter the room.

Processing instructions

A. Transmission Holograms (Direct Process)

N.B. Steps 1- 4 must be done with only the **green safe-light** on.

1. Remove the exposed plate from the 'transport container' and place it, **emulsion side up**, in the tray of **developer**.
2. **Develop for 5 minutes**. The tray should be rocked slowly and **gently** (don't spill any of the developer!) from side-side and top-bottom throughout this time, since the developing chemicals in the vicinity of the emulsion are quickly used up.
3. **Wash** in a tray of water (= stop bath) for **3 minutes**.
4. Place **emulsion side up**, in the tray of fixer, and **fix for 3 minutes**, rocking continuously as before.

You may now turn on the room lights.

5. **Wash** in running water for **5 minutes**.
6. **Bleach (direct)** until clear (approx **1 - 2 minutes**). **Do not discard the bleach**,

return it to the bottle.

7. **Wash** in running water for **10 minutes**.
8. **(Optional)** Wash in 'Agepon' (a weak detergent) for 1 minute to prevent streaks on drying. Dispose of the Agepon down the sink.
9. **Dry** in a dust free environment (you can use a hair drier, gently, on low setting).

B. Reflection Holograms (Reversal Process)

N.B. Steps 1- 4 must be done with only the **green safe-light** on.

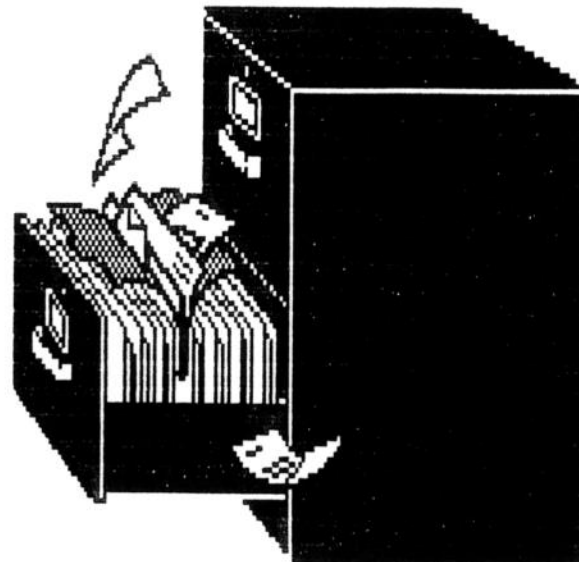
1. Remove the exposed plate from the 'transport container' and place it, **emulsion side up**, in the tray of **developer**.
2. **Develop for 5 minutes**. The tray should be rocked slowly and **gently** (don't spill any of the developer!) from side-side and top-bottom throughout this time, since the developing chemicals near the emulsion are quickly used up.
3. **Wash** in a tray of water (= stop bath) for **3 minutes**.
4. **Bleach (reversal)** until clear (approx **1 - 2 minutes**). Return bleach to the bottle.

You may turn the room lights on after 1 minute of bleaching.

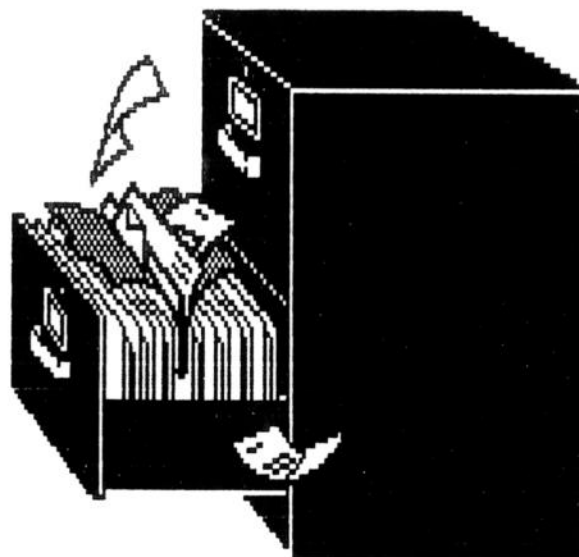
5. **Wash** in running water for **10 minutes**.
6. **(Optional)** wash in 'Agepon' (a weak detergent) for 1 minute to prevent streaks on drying. Dispose of the Agepon down the sink.
7. **Dry** in a dust free environment (you can use a hair drier, gently, on low setting).

Cleaning up

- pour (funnel) the used **bleach back into its bottle**
- pour the **used developer and fixer** into the **liquid waste container**
- thoroughly wash the trays, funnels, tongs etc with water
- wash the sink and clean up any chemicals spilt on the floor etc



SEPARATOR

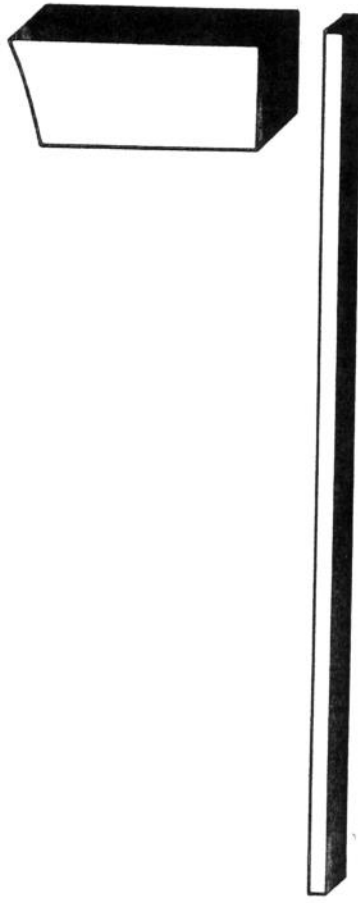


We have included extensive discussions of applications of holography, some of which are in common practice today, and some, such as holographic movies and television, that are still undergoing development.

We have also had in mind readers who may want to try their hands at making holograms. For those readers, we have included descriptions of a number of different ways to make holograms and practical advice about equipping a holographic lab. In doing this, we have stressed the particular method that we have found to be simple, effective, and relatively inexpensive. For those who wish to go further in experimentation, we offer additional assistance.

May readers enjoy this book as we have enjoyed writing it!

This book is dedicated to the students of holography who have used this book while it was being prepared and helped greatly to improve it—and to all persons who will use it in the future.



Holography and ordinary photography

INTRODUCTION

A hologram is a photographic emulsion in which information about a scene is recorded in a very special way. When the hologram is properly illuminated, you, the viewer, see a realistic, three-dimensional representation of the scene. Upon seeing your first hologram, you are certain to feel puzzlement and disbelief. You may place your hand where the scene apparently lies, only to find that nothing tangible is there. You may then look around, trying to learn what the trick or gimmick is. Holograms are indeed provocative; you will not lose the sense of wonder even after having seen many of them.

The trick in holography is in the special manner optical information about a scene is recorded in the emulsion. This is done in such a way that the information is *complete*, so that the scene can be made visible in all its true spatial 3-D aspects with shadows and varying intensities through the scene reproduced realistically. In fact, the word *holography* is derived from Greek roots meaning "complete writing."

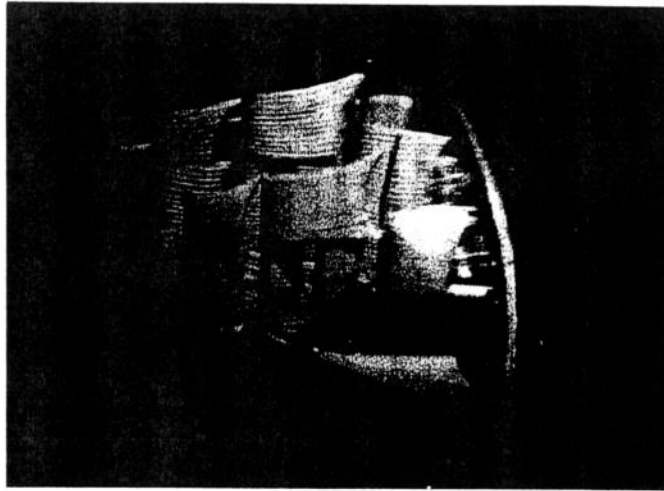
Holograms are not arcane things, shrouded in mystery. On the contrary, they are understandable devices operating on the basis of simple optical principles. This book explains those principles and shows how they account for the properties of holograms. This first chapter discusses some general aspects of holography that should be known before you begin to study the underlying optics in Chapter 2.

The Holographic Scene

One way to view a holographic scene is shown in Figure 1.1. The hologram itself is a sheet of exposed and developed photographic emulsion. The light from a laser in the background shines onto the plate. While on the near side of the plate you look *through* the plate and perceive objects in the scene very much as though you were looking through a window at an actual scene lying beyond the window.

A flat picture such as Figure 1.1 cannot show how striking the real view is. It is immediately obvious that you are seeing something very special. The objects in the scene have *depth*—a three-dimensional appearance. As you move your head, you can see around objects and can see things that from other points of view lie behind other objects. Shadows change their positions. Light seems to bounce off objects at varying angles. A diamond ring in a holographic

Figure 1.1. The virtual image produced by a transmission hologram, illuminated from behind by a laser. Of course, this photograph cannot convey the vivid three-dimensional nature of the actual scene. (Photo by Dr. C. E. Hamilton, produced by Mr. Rainbows.)



scene can sparkle and show changing aspects of its facets. (One of the first holograms made by the wife of one of the authors was of her engagement ring. It was a striking hologram.)

The particular kind of hologram in Figure 1.1 is called a *transmission hologram* because the laser light is transmitted through the hologram to the viewer's eyes. Another principal kind, the *reflection hologram*, is viewed in light reflected from a light source on the viewer's side into his or her eyes.

The Hologram Itself

The hologram is a layer of photographic emulsion, either on clear flexible plastic material or on a glass backing.¹ For simplicity we will refer to the hologram as a *plate*. Inspection of the plate reveals a remarkable fact. There is nothing on it that resembles the scene. The plate may be quite clear or somewhat cloudy, or it may have dark swirls and striations.

When you look at a hologram such as Figure 1.2 the obvious question is, "Where is the scene?" In fact, the scene is not there at all; rather, it is

Figure 1.2. The appearance of a hologram plate (unbleached). The swirls and striations seen are not the microscopic interference patterns that really constitute the hologram. They are due to dust or imperfections in the optical apparatus that was used to make the hologram. (Photo by Dr. C. E. Hamilton.)



¹In recent times substances other than the familiar silver halide photographic emulsions have been developed for making holograms.

information about the scene, coded in the form of *interference patterns*, that is recorded in the hologram. The coded information itself would have no discernible resemblance to the scene even if you could see it with the naked eye. Actually, the interference patterns are present on a microscopic scale. If you happen to see swirls or other markings as in Figure 1.2, these are caused by extraneous effects such as dust particles or imperfections in some optical apparatus used in making the hologram.

It is only when the hologram is suitably illuminated that the information contained in the hologram can be decoded and the scene *reconstructed* or made visible.

MAKING PHOTOGRAPHS AND HOLOGRAMS

It must be a rare person who does not know something about the making of ordinary photographs. Nevertheless, we outline the process here. In Figure 1.3 a camera is used to photograph an object (a simple cylinder). There is a source of light, and some of the light falls on the object and reflects into the lens of the camera. The lens focuses the light onto photographic emulsion so that a sharp image will result. A shutter (not shown) controls the length of time of exposure of the emulsion. After exposure, the film is removed from the camera and is processed. The immediate result is a negative from which a positive print can be made.

Figures 1.4 and 1.5 show one arrangement for making a hologram. Let us first describe the configuration in general and then in more detail.

As in ordinary photography, there must be a source of light to illuminate the scene and activate the photographic emulsion. The light source shown is

Figure 1.3. Making an ordinary photograph.

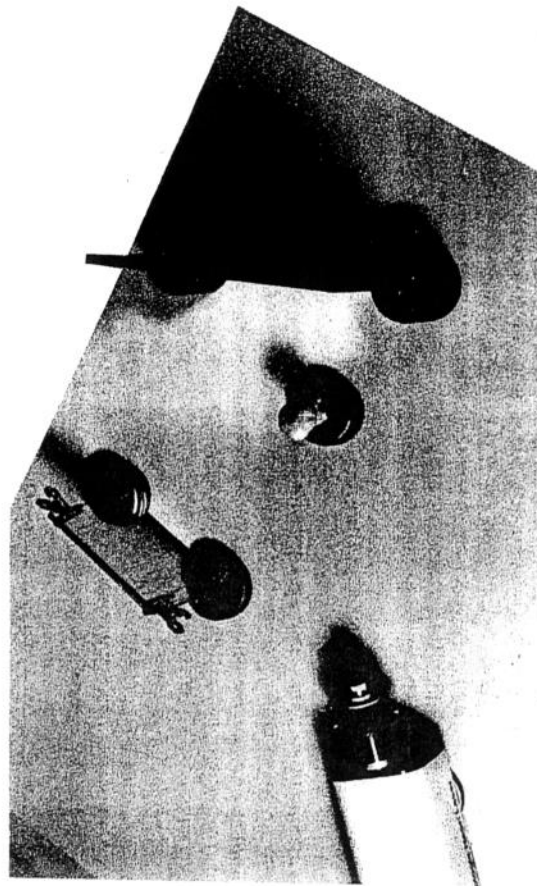
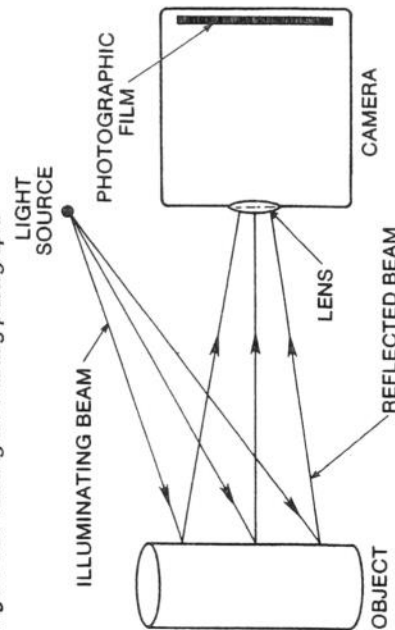
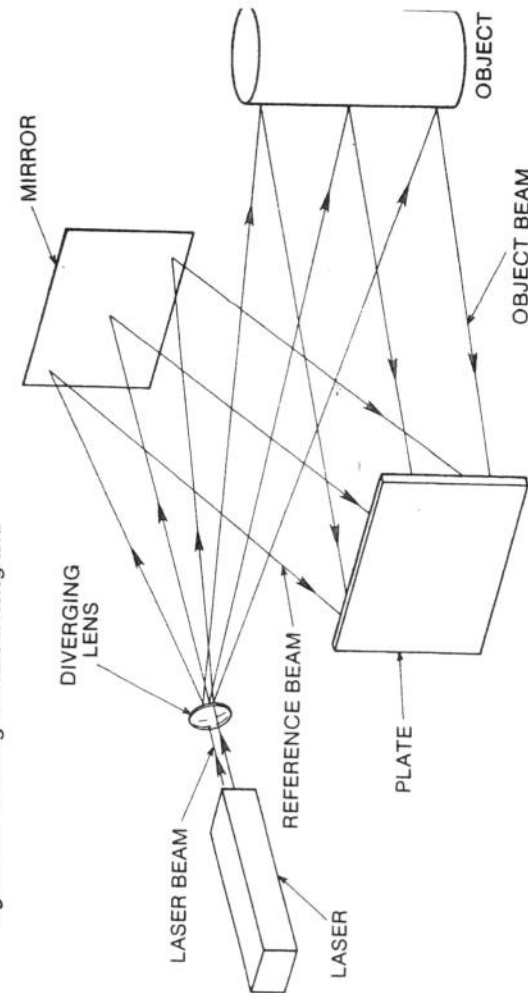


Figure 1.4. Set-up of apparatus on a holographic table top for making a one-beam transmission hologram. Laser at lower left; front surface mirror at right; film holder containing film at upper left; object (a small doll) in center. In this compact arrangement a strongly diverging lens is used at the port of the laser. More commonly the laser is placed farther off to the left. (Photo by Dr. C. E. Hamilton.)

Figure 1.5. One way to make a hologram.



a laser, for reasons that will be made clear. The laser beam is narrow, only a few millimeters across, and must be spread out so that it can illuminate substantial areas. The divergence of the beam is accomplished by a lens.

Now something of great importance is done. Some of the spread-out beam of laser light falls on a flat mirror that is oriented so that the light reflecting from it hits the plate, while another part of the beam bypasses the mirror, falls on the object, and reflects back to the plate. These are the *reference* and *object* beams, respectively. When the shutter (not shown) is opened to expose the plate, these two beams of laser light interact with each other. An interference pattern is formed and is recorded in the emulsion. After the plate is processed, you have a permanent record in the form of a negative of the interference pattern. This is a hologram.

Before proceeding to the fundamentally important aspects of this scheme, two simpler matters deserve comment. In making a photograph with a camera, the lens is essential in getting an image of the scene. In contrast, the lens shown in the arrangement in Figure 1.5 is not used in that manner at all. It is there only to spread out the laser beam. You may hear it said that holography is "lensless," and this is true in the sense that the lens is not used for focusing the image. Second, special film must be used in holography because the interference pattern to be recorded in it is on a microscopic level of fineness. Photographers would call this *very high resolution film*.

COHERENCE AND THE LASER

In making a photograph, as in Figure 1.3, there is nothing special required of the light source. An ordinary light bulb will serve, or several bulbs, as might be used in an interior room. Sunlight or candlelight can be used.

In contrast, the holographic method *always* requires that two special conditions be met. One is that there must be specifically *two* beams incident on the plate—the reference beam and the object beam. The two beams are required to form the interference pattern.

The other special property of light used in holography is that it must be *coherent*. We will have more to say about this later, but you will be helped considerably by a brief explanation now. This will also make it possible to explain the role of the laser in holography.

Let us begin by saying that you cannot make a hologram with a flashlight, the sun, or any ordinary source of light. What we call "ordinary" sources produce light that lacks coordination between parts of the light beam. They are said to be *incoherent* sources.

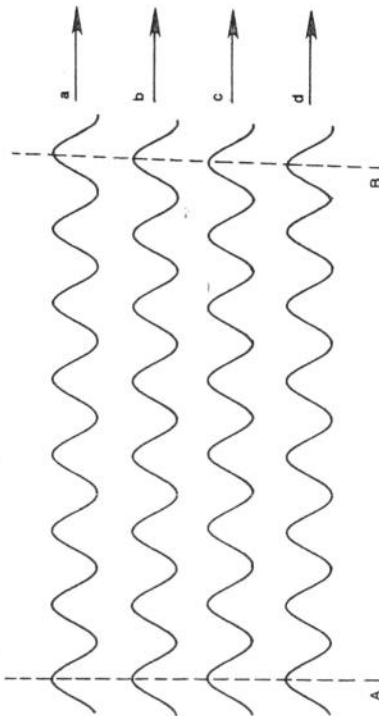
The difference between coherence and incoherence can be explained by a simple analogy. Imagine a body of troops marching in orderly ranks and files, with every soldier perfectly in step with every other soldier. In optical terminology this would be considered a *coherent* formation. If, on the other hand, there

were no synchronization between one soldier's pacing and that of another, the formation would be *incoherent*. Now consider a beam of light. The light consists of electromagnetic waves. Corresponding to a file of marching soldiers there is an oscillation in the beam along the direction of travel, and corresponding to the ranks of soldiers there are oscillations across the breadth of the beam. This is illustrated in Figure 1.6. If all the oscillations are perfectly regular and all are in step with all the others, the light is coherent. In the case of incoherent light, such synchronization among parts of the light beam is not present.

Why is it necessary to use coherent light in making holograms? Suppose that two beams of coherent light are superimposed, as are the reference beam and the object beam in Figure 1.5. They interact with each other in the plate, producing a very finely detailed pattern of exposed and unexposed regions, and this is the hologram. If at some point the two light beams are completely in step with each other throughout the exposure, there will be a maximum blackening. If at some point the two beams are completely out of step with each other during the exposure, they will cancel out each other's effect and there will be no darkening. In this way, alternating light and dark regions (interference fringes) are formed. This is the desired result, but it would be spoiled if the requirement of coherence were not met. Suppose that at some point the light beams were sometimes in step and sometimes out of step, with random fluctuations in the relationship. If this situation extended over all of the emulsion, the processed plate would show only a darkening, and no hologram would be produced.

Where can you get the coherent light required? At the present time, there is only one satisfactory source: the laser. A laser is a device that produces a

Figure 1.6. Schematic representation of a beam of fully coherent light. The arrows (a, b, c, d) indicate direction of travel. Along each "file" the waveform has its peaks and valleys uniformly spaced. Along each "rank" (as along A and B) the waves are in step.



highly coherent beam of light.² Fortunately, the common helium-neon (He-Ne) gas laser can now be obtained with sufficient power at a cost as low as about \$300—and it is simple to operate, long-lived, and safe. Most holograms today are made with the He-Ne laser.

A laser beam has two other special characteristics. One of these is that the light is highly *monochromatic*, which means that the color is very sharply defined. In more technical terms, the light is very nearly of one single wavelength. For the He-Ne gas laser the color is a red that resembles that used in neon advertising signs. It happens that the coherence and the monochromaticity of the laser beam are related. This will be explained later.

The other special property of a laser beam is that it is highly collimated. It issues from the laser with a small cross-sectional dimension, usually a few millimeters. As the beam travels along, this dimension increases very little, even over very great distances. Although the coherence and purity of color of the laser light are very important in holography, the collimation is an obstacle. As we have seen, a setup for making a hologram begins by spreading the beam out with a lens.

There are those who think every laser beam is extremely intense. Actually, the beam may be only as intense as that from a flashlight, intense enough to vaporize metal, or anything in between those extremes. The only importance of the intensity of the laser beam in holography is that it be great enough to permit exposure time of the photographic emulsion to be reasonably short, and low enough to do no damage to the objects (which might be people!) being holographed.³

VIEWING PHOTOGRAPHS AND HOLOGRAMS

Figure 1.7 represents an eye looking at an ordinary photograph. Light from an ordinary source such as the sun or a room lamp casts light onto the surface of the photograph, some of the light reflects off into the eye, and the lens in the eye focuses the light onto the retina at the back of the eye. Then a very complicated physiological system takes over, with the result that the viewer perceives the picture.

What you see is an intrinsically flat surface. Only by complicated mental processes do you relate what you see to the original scene.

You are probably familiar with the stereoscope in which two side-by-side photographs are viewed simultaneously, or with 3-D television in which colored glasses are used to give a different image to each eye. These depend on the phenomenon of "binocular vision" and give some sense of depth in the scene,

²Even a laser does not produce perfectly coherent light. There can be excellent coherence across the beam but inevitably only over a limited extent along the beam. Chapter 3 explains this further.

³See "Laser Safety" in Appendix I.

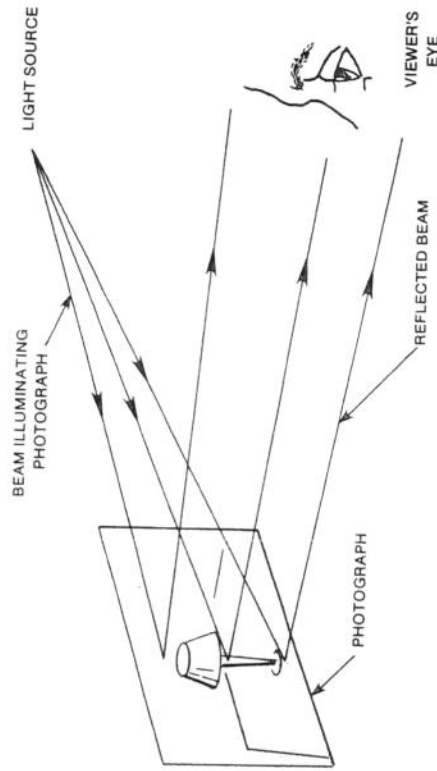


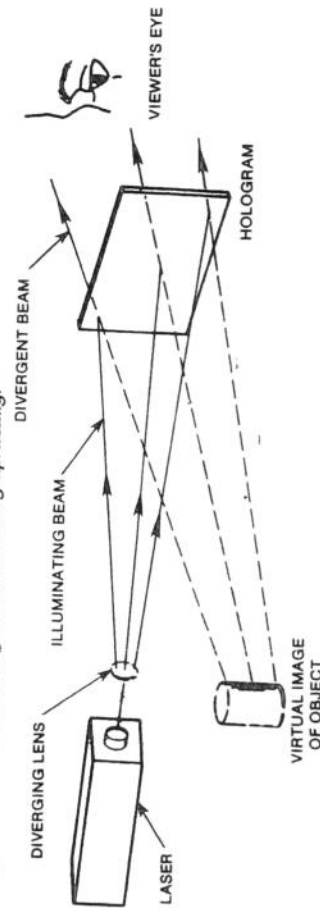
Figure 1.7. Viewing an ordinary photograph.

but they do not meet the acid test for three-dimensional realism: Can you shift your point of view and look around a tree, for example, and see objects that are behind it?

Now let us consider Figure 1.8 which is a diagrammatic version of Figure 1.1. It shows how to view the scene with a transmission hologram.⁴ The laser beam is diverged by a lens to spread it enough to illuminate the plate. Your eye picks up the light on the other side of the hologram, after it has been acted upon or "decoded." You can look around the tree.

A hologram differs from a photograph in a number of striking ways other

Figure 1.8. Reconstructing the scene holographically.



⁴If the hologram were of the reflection type, the arrangement for viewing would resemble that in Figure 1.8, but the light source would be on the viewer's side, as it would be if he or she were looking at a mirror.

than the realism of the image it provides. One of these is that while a photograph is a single representation of a scene, a hologram gives more than one image. Figure 1.9 provides a basis for this discussion. The hologram is a transmission hologram illuminated by a diverged laser beam. Some of the laser light passing through the plate is acted upon by the interference pattern recorded so that it emerges as a diverging beam. This is suggested by the rays labeled A, B, and C. When your eye receives this light, the lens in the eye is used to focus the light on the retina, and you interpret what you see as though the light came from a nonexistent *virtual* source. The eye reacts as though the rays A, B, and C were extensions of the fictitious rays A', B', and C'. These fictitious rays would exist if there were a real object where the "virtual image" is shown. The object is "seen" as if it were at that location, even though there is no real object there.

Actually, looking at a virtual image is an everyday experience for most people and evokes no mystery. When you look into a mirror, you see an image of yourself as though there were a person behind the mirror while in reality the mirror is opaque (though reflecting). Light from a source on your side of the mirror reflects into your eyes and you interpret what your eye receives as a "virtual image."

Some of the laser light comes from the hologram as a *converging* beam. This is suggested by the rays D, E, and F in the drawing. These come to a focus and produce a second image called the *real* image. This means that the light (rays D, E, F and others) converges, focuses, and creates an optical replica of the object. If you were to put a piece of paper there, you could look at the paper and see the image. No focusing lens is needed to help the rays produce this image, unlike the case of the virtual image. Since the image is really there, it can be viewed with the eye, as when looking at a photograph.

Like the virtual image, the real image is fully three-dimensional. A piece of paper put there to reveal it cannot show you all parts of the image in focus. Instead, you will see only a slice through the image and, as the paper is moved alternately closer to and farther from the hologram, various parts of the scene go into and out of focus.

If you have a transmission hologram, a laser, and a suitable lens at hand, you can see the virtual image by the arrangement shown in Figure 1.9, and by searching with a piece of paper you should be able to find the real image as well. (This is discussed further in Chapter 5.)

We turn now to another fundamental property of holograms. It is called *redundancy*. If you were to cut a photograph into pieces, you would not be able to look at any single piece and see the whole scene. You would have only pieces of a jigsaw puzzle that would have to be reassembled properly to reproduce the scene. On the other hand, if you were to break a hologram into pieces or, more practically, use a card with a hole in it to block off all but a small segment of the hologram, you would still get a realistic three-dimensional view of the scene from the segment. You can correctly think of viewing a holographic

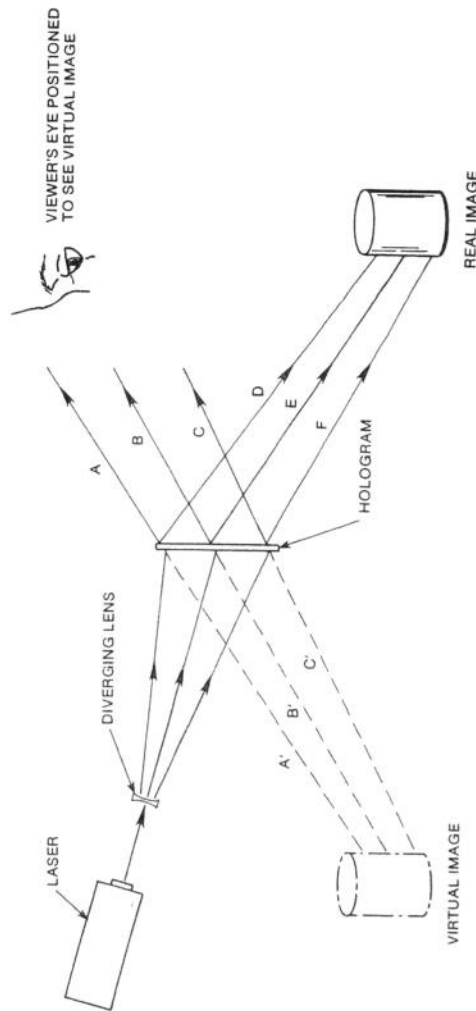


Figure 1.9. The hologram produces two images.

scene as like looking through a window. If you use only a small piece of a hologram, this is like looking through a small window, while using the whole hologram is like using a larger window. You would be limited in moving your head around to view the scene, but what you see would otherwise be fully realistic. Thus a hologram records information about the whole scene in each small piece of the plate.

Attempts to understand how the human brain stores information have heretofore been based on the assumption that it is a localized process, with each piece of information stored in a definite place, as in the memory of a computer. There are difficulties with this idea; removal of part of a brain does not necessarily remove a definite portion of memory. It has been suggested that the human memory is perhaps like a hologram in that what is stored is spread out over whole regions, with a small portion of the brain able to retain the information.

CONCLUDING REMARKS

We have tried to give you a general idea of what holograms are, how they are made, and how they are used to reconstruct the scene. There are other kinds of holograms, other techniques for making them, and other interesting properties inherent in them. However, these matters are best discussed after you learn more about wave theory, which begins in the next chapter.

institution, the physics department probably has a mercury vapor lamp. Such a lamp emits light that can be filtered to give reasonably intense and monochromatic blue light. In fact, the world's first holograms were produced by Dr. Gabor with such a source long before lasers existed.



Holography: The geometric model

INTRODUCTION

The physics of holography is well understood today. The complete theory is, however, highly mathematical. It begins by representing light waves with certain abstract expressions and then explores mathematically the interference and other phenomena that arise.

Fortunately, it is possible to extract from the elaborate mathematics of the theory intuitively understandable views of holography. The purpose of such models is to offer simple ways to think pictorially of phenomena, though with sacrifice of completeness.

In fact, two good models of holography have been devised—good because they are easy to grasp and because they give insight into how holograms work. They are so useful even specialists in holography rely heavily on them.

One of these is the *zone plate* model. This model has been used in popular and technical discussions of holography since the time of Dr. Gabor's original invention of the subject. It is a powerful model and easily understood once you have learned about a class of optical effects known as "diffraction" phenomena.

An even simpler model of holography was devised by Dr. Tung H. Jeong

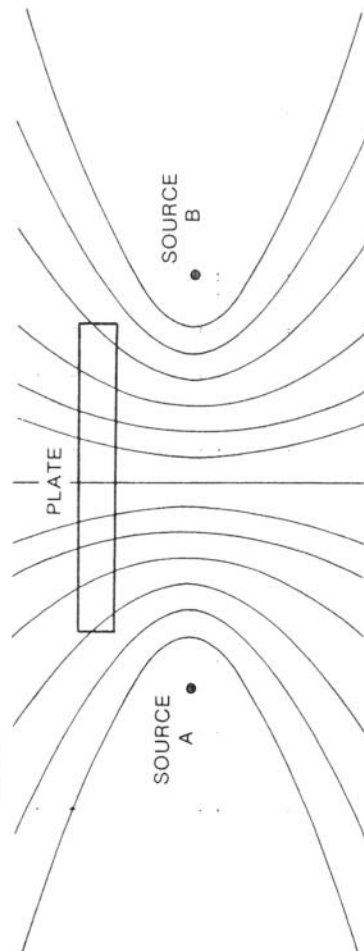
of Lake Forest College in Illinois.¹ For many years Dr. Jeong has been a leader in holography. His *geometric model* is especially easy to understand, though limited in certain ways in its applicability. Because it is so clear and so effective where it does apply, the geometric model will be discussed first, leaving the zone plate model for a later chapter.

PHOTOGRAPHING THE INTERFERENCE PATTERNS

Chapter 3 discussed how light waves from two coherent sources interfere. The interference produces a stationary pattern of constructive and destructive *fringes*. This is shown again in Figure 4.1. The role of the photographic plate shown will be examined later.

The fringes as shown in the sketch are cross-sections of what are actually three-dimensional surfaces.² You can visualize the surfaces by thinking of rotating Figure 4.1 about a line through A and B. For clarity, part of one of the surfaces is shown in Figure 4.2, again intercepted by a plate. At all points on this surface the intensity of the light is at its greatest due to constructive interference. The next neighboring surface (not shown) also consists of points where there is constructive interference, and between the surfaces there is low intensity of light, due to destructive interference.

Figure 4.1. For making a transmission hologram a photographic plate has been placed in the interference pattern that is produced by the coherent sources A and B.



¹Tung H. Jeong, "Geometric Model for Holography," *American Journal of Physics*, August 1975, pages 714-717.

²As a matter of terminology, the three-dimensional surfaces are hyperboloids or, more properly, hyperboloidal surfaces, although they are often simply called hyperbolas.

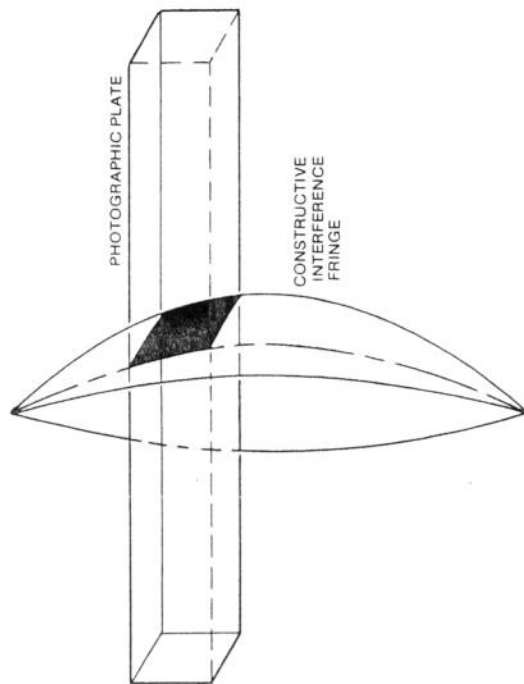


Figure 4.2. Part of one of the hyperboloidal surfaces in the interference pattern. The segment shown as dark activates the corresponding part of the emulsion.

An essential step in holography is producing the stationary fringe pattern. The next step is recording the pattern to form the hologram. The emulsion in Figures 4.1 and 4.2 reacts strongly where the light is intense and less strongly elsewhere. Thus the constructive interference fringes will become surfaces where there has been strong activation of the silver halide grains in the emulsion. After the exposed plate has been processed chemically, it will be a photographic negative that contains a permanent record of the interference pattern.

The next step in holography is to use the plate to reconstruct a visible decoded version of the information that was recorded. Before discussing this step, let's consider certain properties of the hyperboloidal surfaces further.

THE FRINGES AS REFLECTING SURFACES

Figure 4.3 shows the nature of the photographic negative in a cross-sectional view. Each of the fringes is now to be thought of as a *partially reflecting surface*. Suppose you were to shine light on the plate. Where the incident light falls on one of the surfaces, some of it will be reflected back, some will be transmitted through, and some will be lost by absorption. The general nature of the reflection from one of the surfaces is shown in Figure 4.4. Light from the real source A

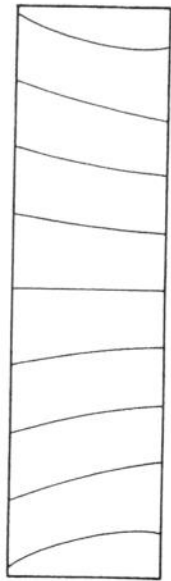


Figure 4.3. The nature of the interference fringes that are recorded in the plate.

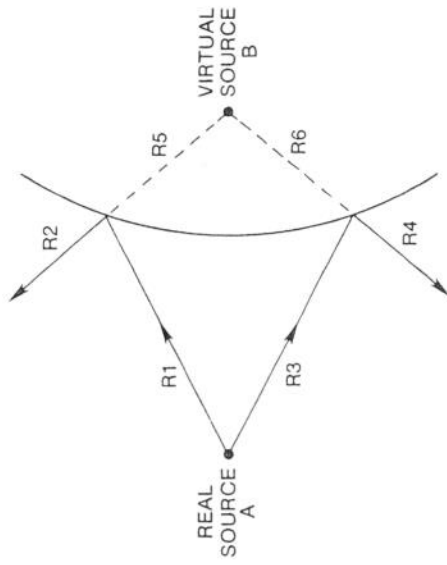
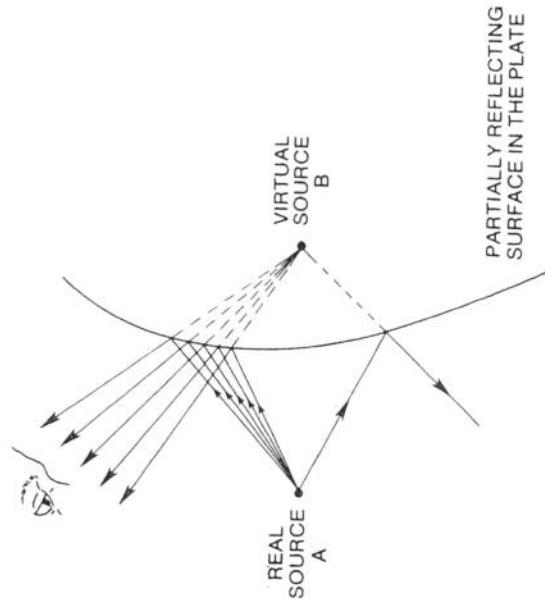
reflects in the manner shown to the observer's eye. (The "virtual source B" will be discussed presently.)

The whole plate is to be thought of as a set of nested partially reflecting surfaces, each of them acting like the one in Figure 4.4 when the plate is illuminated.

The importance of the hyperbolic surfaces lies in the precise way in which they reflect light rays. Consider Figure 4.5 (a). The curve shown represents one of the partially reflecting surfaces, originally made using real sources A and B. Now the source B has been removed, and only the source A illuminates the plate. The ray R_1 reflects from the surface as ray R_2 . Similarly ray R_3 reflects as

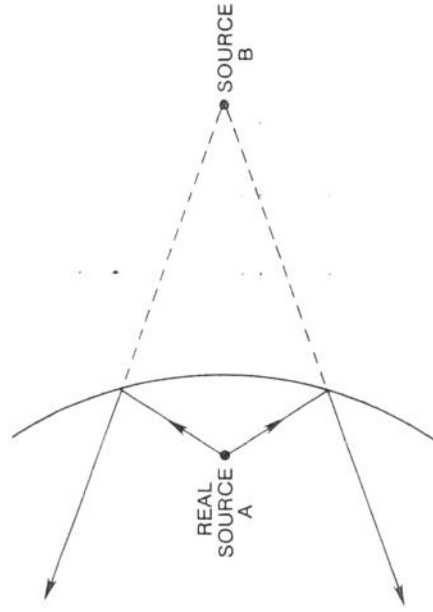
Figure 4.4. The manner in which rays from a real source of light (A) reflect from one of the partially reflecting surfaces in the plate.

OBSERVER'S EYE



(a)

Figure 4.5. Rays from the point source A seem after reflection to have come from a virtual point source B. (a) Convex surface. (b) Concave surface.



(b)

ray R_4 . If the reflected rays R_2 and R_4 are traced backward, as with the dashed lines R_5 and R_6 , these extensions will intersect at the point B. In part (a) of the figure, the surface is convex toward A. The theorem holds equally well in the case shown in part (b), where the surface is concave toward A.

Furthermore, this holds for any rays from A. Hence in Figure 4.4, all of the rays shown as emanating from the source A and reflecting from the surface will, when traced backward from their reflected parts, intersect at B.

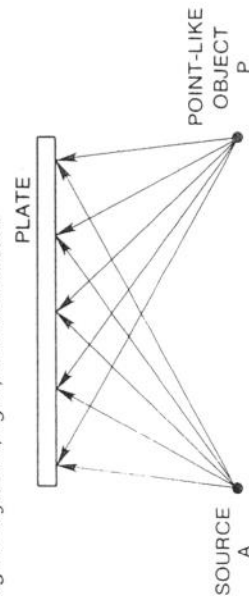
If you see the reflected rays as in Figure 4.4, what will you perceive? The eye has no way of knowing that the light rays received really came from the source A after reflections and that there are no real rays emanating from the point B. To the eye, it is precisely as though the rays *did* come from B. This is called a *virtual image*, thereby emphasizing that the source is not really there. As pointed out in Chapter 1, virtual images are far from being esoteric optical entities; they are seen whenever you look into an ordinary mirror.

A HOLOGRAM OF A POINTLIKE OBJECT

At last we are ready to discuss making a hologram of something and viewing it. We do this in two stages. In this section the "something" to be holographed will be only a very small, almost pointlike object. In the next section, the ideas that will have been developed will be applied to the general case—a hologram of an extended scene.

In Figure 4.6 some of the light from source A goes directly to the plate; this is the reference beam. Some of the light (not indicated by rays in the drawing) goes from the source A to the object P. This reflects from P, and some of it falls on the plate. The latter light is the object beam. The reference and object beams interfere in the emulsion. After the plate is exposed and processed, it is a hologram of the pointlike object.

Figure 4.6. Making a transmission hologram of a pointlike object P. The rays from the light source A that go directly to the plate constitute the reference beam. The object beam consists of the rays from P to the plate. These are created by reflection by the object P of light from the source A.

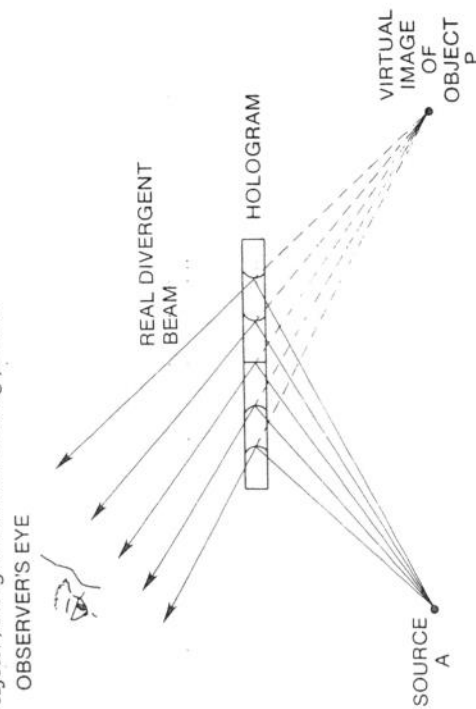


It would be well to compare Figures 4.1 and 4.6 at this point. Although in 4.1 the source B is self-luminous, while in 4.6 the object P shines only by reflected light, yet the geometry is the same in the two cases. That means that the *shapes* of the fringes and their *locations* in the emulsions are the same in the two cases. Hence with reconstruction as in Figure 4.7 a pointlike virtual image will be seen in either case, and in the same apparent location. However, how the virtual image looks depends on still another property of the fringes—their degrees of blackness or their contrast with the generally more transparent background in the emulsion. If the reflecting object P is so shiny that it reflects almost all of the light that falls on it, the interference of the reference and object beams in the emulsion will produce fringes with high contrast. When the image is reconstructed, it will be bright. On the other hand, if the original object P is dull, reflecting little light, the fringes in the emulsion will be of low contrast, and the image will be dull.

This is of fundamental importance, for it is the basis for understanding how the holographic image of a scene can show varying degrees of brightness in various parts of the scene, as is required for a realistic representation of the original scene.

Besides this matter of "dynamic range," the realism that is perceived in the reconstructed image depends on having all the points in the scene in their proper relative positions in three-dimensional space.

Figure 4.7. Viewing the hologram of a pointlike object. The reconstruction with light from the source A gives the viewer the impression of seeing the object P, though this is not actually present.



HOLOGRAM OF AN EXTENDED SCENE

If, in making the hologram as in Figure 4.6, the object P had been located at a different point (nearer to or farther from the plate or displaced in some other fashion), the fringes recorded in the hologram would be different from before and in a manner corresponding to the displacement. On reconstruction of the virtual image as in Figure 4.7, the viewer would perceive the object as though it were located where it was originally. Thus the plate contains information about position in the three-dimensional space.

Figure 4.8 shows how a transmission hologram of a sizable object can be made. The coherent light needed is a beam from a laser. The beam is diverged by a lens to spread it out enough to illuminate the object. In the figure, four points on the surface are shown as reflecting light. Sets of hyperbolic surfaces will be formed by interference, one for each part of the surface. Each set will differ in detail from the others.

When the hologram is viewed, the four parts of the surface will be seen, each in its proper spatial position with respect to the others and each with its proper brightness or dimness. In short, they will be seen recreated realistically.

At this point a mental leap is required. You are to think of all the points of the object as behaving in the same way as our four representative points. The hologram of the object will contain a very complicated collection of sets of interference fringes. However, when the hologram is illuminated for viewing, the inexorable laws of reflection of light rays take over. The various sets of fringes will produce images of the different points, all properly sorted and placed in their correct spatial relationships. The scene will have true depth.

The arrangement for reconstructing the scene is illustrated in Figure 4.9. (No attempt has been made to cope with the enormous complexity of the fringes in the real situation.) Notice that if you put your eye to the left in the "viewer's location" region, you will see the object from a corresponding direction and similarly if you move to the right. In short, you will be able to look part way around the object. Suppose there were two objects used in making the hologram, one partly behind the other. On reconstructing the scene, you could, by moving your head, look part way around the image of the nearer object and see parts of the farther object come into view.

PHASE AND AMPLITUDE INFORMATION

The most fundamental characteristic of a hologram is that it records both *phase* and *amplitude* information. In contrast, an ordinary photograph records only amplitude information.

In order to understand this much used but highly succinct statement, you must first understand what is meant by phase. Imagine two light waves traveling side by side, having the same wavelengths and the same amplitudes as any

two of those shown in Figure 3.4. All of those shown in the drawing are also in step across the beam. Now concentrate on the uppermost two of the waves and think of the second of them as shifted to the right. After this shift, the two waves are still identical with respect to wavelength and amplitude (or color and intensity), but they differ in *phase*. To become more precise about this, Figure 4.10 shows five waves, all the same except for phase shifts relative to each other. Take A to be one with respect to which the shifts of the others can be measured. One way to specify the amount of shift of any of the others is to state it as a fraction of one wavelength of the light. This will be the *phase* of the wave relative to the standard A. The phase of B is 0 and the phases of C, D, and E are approximately $1/8$, $1/4$, and $1/2$ wavelength, respectively.³

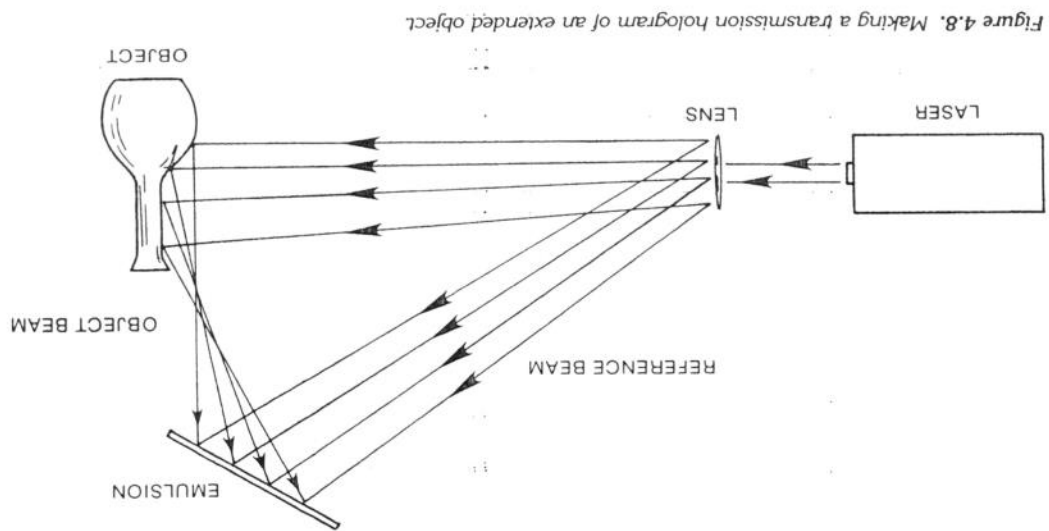
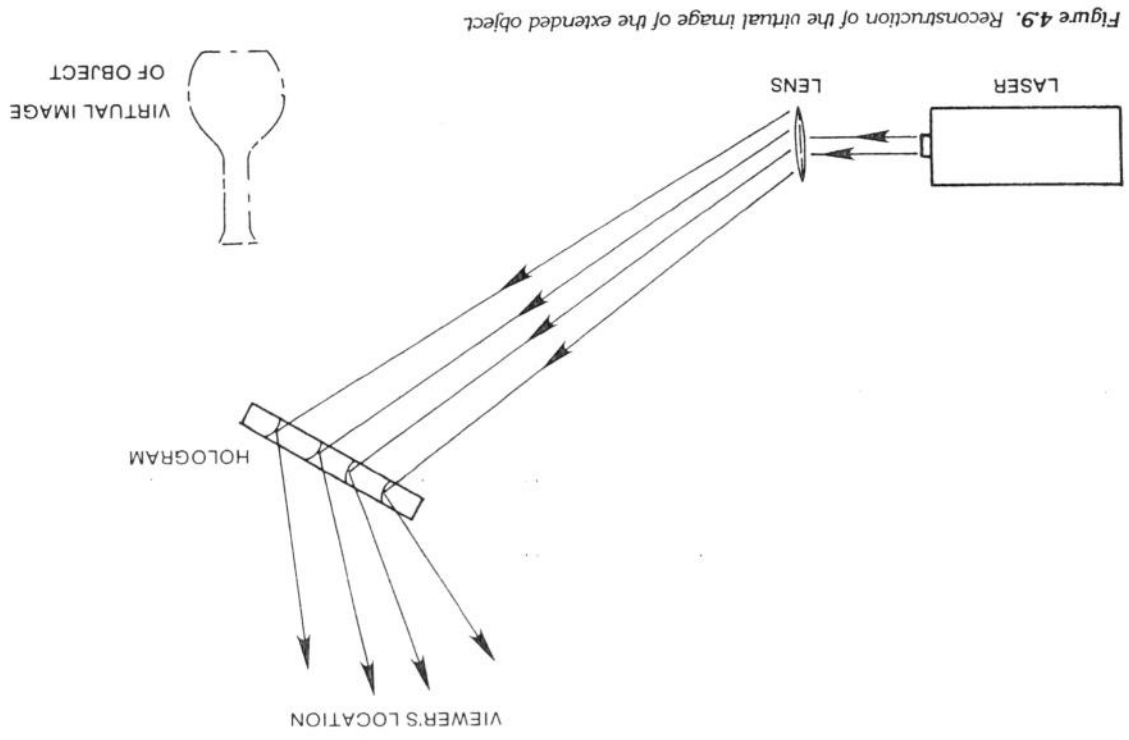
In making a hologram, a reference beam and an object beam interfere. For constructive interference, the beams are said to have zero relative phase, or that they are *in phase*. For destructive interference, the beams have relative phase $1/2$ wavelength. They are completely *out of phase*. For intermediate phase differences, the interference is partial. What is recorded in the photographic emulsion is the interference pattern, and this depends at every point upon the *phase relationship* between the reference and object beams. Therefore, when the emulsion records an interference pattern, it is truly recording phase information.

What about amplitude information *from the scene*? This question can be distinguished by the way in which phase information is converted into amplitude information when the interference occurs in the plate, and by amplitude information about the scene carried by the object beam itself, before the object beam encounters the reference beam in the plate. After all, if you look at something in ordinary light, you detect differences in brightness in various parts of the scene, and this is not a matter of interference with a reference beam. Amplitude information about the scene in this sense must also be recorded in a hologram. In short, how does the hologram know what parts of the scene are bright and which are dim?

In making a hologram as in Figure 4.8, the various parts of the object being holographed have various reflectivities, various intensities of light falling on them, and are at various distances from the plate. All of these factors contribute to determining how intense the object beam is when it meets the reference beam at each point in the plate. A weak object beam somewhere means weak interference, and a strong object beam elsewhere means strong interference there. Thus the degree of blackening of an interference fringe in the plate depends on the amplitudes of light from various parts of the object as well as on phase. A hologram records this information and so can be said to contain amplitude information about the scene.

In making an ordinary photograph (with ordinary light, of course), there

³Another way is to consider one wavelength to be equivalent to 360 degrees. With this convention, the phases of B, C, D, and E relative to A are approximately 0, 45, 90 and 180 degrees.



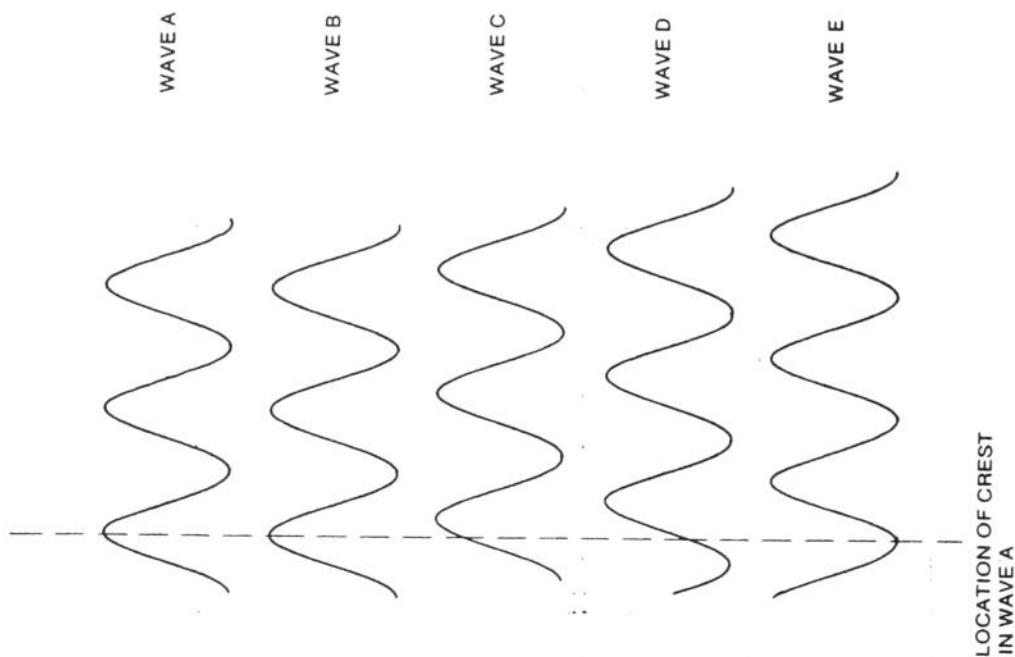


Figure 4.10. Using wave A as a reference wave, the phases of the other waves can be specified quantitatively.

is no possibility of recording phase information. Incoherence smears out the interference fringes and only amplitude information is left. That is, bright spots in the scene activate the emulsion in corresponding parts more than do dimmer spots in the scene, and this is purely amplitude information.



Figure 4.11. The general nature of a portion of a hologram, as it would appear under a microscope.

THE REALITY OF THE PARTIALLY REFLECTING SURFACES

In using the geometric model, you may think of the hyperboloids as partially reflecting surfaces. At the same time, it is going too far to take the model so literally as to suppose that the surfaces are continuous, shiny surfaces like our familiar silvered mirrors. For one thing, photographic emulsions are necessarily grainy, and this graininess limits their ability to contain continuous surfaces in them. Also, when a scene is holographed, the exceedingly numerous points in the scene produce innumerable overlapping fringe systems, with a resultant complexity so great that even in the absence of the graininess effect a microscopic view of the plate could hardly be expected to reveal distinct continuous surfaces. In fact, a hologram seen under a microscope appears as a collection of squiggles, somewhat as in Figure 4.11.

That this is so may seem to make it miraculous that the geometric model works at all. But if the pieces of mirrors (the squiggles in the hologram) have sufficient organization among them, as they do, various sets of pieces can act consistently as fragments of continuous mirrors. As an analogy, you might think of a mosaic of pieces of silvered mirrors embedded in plaster and so cunningly put into place that from various viewpoints you can see your face reflected.

REFLECTION HOLOGRAMS

A transmission hologram is viewed with the light source and the eye on opposite sides of the plate, so that the light is transmitted through the plate. A reflection hologram is viewed with the light source on the same side of the plate as the observer's eye. The light reflects from the plate to the eye, thus the name is appropriate. Since every hologram is illuminated from the one side or the other, every hologram is of one kind or the other. (However, there are numerous varieties of each kind, as you will see later.)

As a first step toward understanding reflection holograms, consider the

arrangement shown in Figure 4.12. A and B are again point sources of coherent light and they produce the hyperbolic interference fringes. The new feature in the arrangement is that the photographic plate is oriented differently from how it would be if a transmission hologram were being made. The plate now has the interference fringes extending essentially lengthwise along the long dimension of the plate. The nature of the hologram formed is represented schematically in part (b) of the figure.

As a second step, suppose that point A is a source of coherent light, but that B is a small (pointlike) reflective object. The plate will receive a reference beam from A and an object beam of reflected light from B. A hologram of the object will be formed in the plate, for the emulsion will react to the varying

Figure 4.12. The plate is placed in a position for making a reflection hologram. (a) The interference fringes lie essentially along the length of the plate. (b) The general nature of the fringes ultimately recorded in the plate.

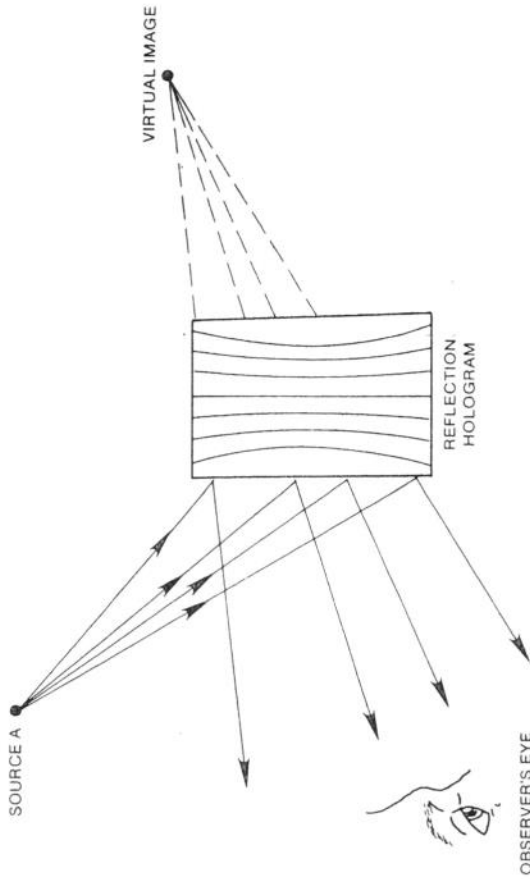
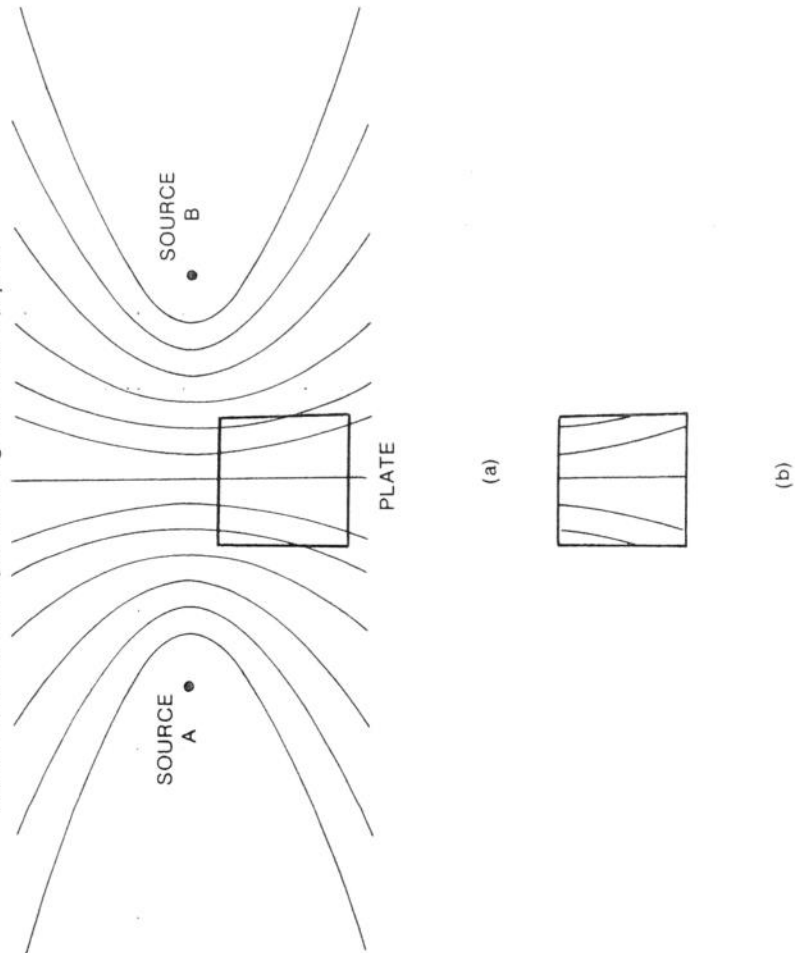


Figure 4.13. Reconstructing the image with a reflection hologram. The light source and the viewer's eye are on the same side of the plate. For simplicity the rays are shown as if they reflected from the face of the plate, but in reality they reflect from the surfaces that are shown in the plate.

intensities of light in the interference pattern. The geometry of the fringes recorded in the hologram will be different from a transmission hologram, however.

As a third step, observe that the geometric model provides for reconstruction of the virtual image of the small object. This is illustrated in Figure 4.13. Light from the source A reflects from the surfaces in the hologram and back to the eye. You interpret what you see as though the light rays came from an object at B.

As a last step, imagine the point B as but one point of the very many points that make up the surface of an extended object. Every point will give rise to fringes in the hologram so that when the hologram is viewed each point will be perceived in its proper three-dimensional relation to all of the others and with appropriate brightness. The viewer will interpret the ensemble of the impressions received as a realistic virtual image of the object.

THE STRATEGY FOR THE NEXT THREE CHAPTERS

In this chapter the geometric model has been used to explain much about transmission holograms but our story is incomplete. For one thing, only virtual images, not real images have been discussed. The latter topic is somewhat

elaborate and will be taken up in Chapter 5. A number of other properties of transmission holograms will also be examined.

This chapter also introduced reflection holograms, but there is much more to be said. In particular, our introduction did not deal with an optical effect called Bragg reflection, which is predominant in reflection holograms. This is very important, especially in connection with color effects. Since it requires some preliminary delving into optical theory, Chapter 6 takes this up, along with the general topic of reflection holograms.

Chapter 7 uses the enlarged theory (geometric model plus Bragg reflection) to examine both transmission and reflection holograms further.



More about transmission holograms

INTRODUCTION

Chapter 4 explained in an introductory way how the geometric model is used to understand the formation of images in holograms. This chapter applies the model to transmission holograms in particular. You will see that many of their specific properties can be explained simply now.

There is a highly important exception—color effects. Wouldn't it be wonderful if transmission holograms could be viewed with ordinary white light sources such as sunlight or electric lamps, doing away with the need for a laser? Wouldn't it also be wonderful if the images were produced in full natural color? Unfortunately, accomplishing these things is not yet in the realm of practical reality.¹ For example, if a transmission hologram is viewed in ordinary white light, the image is seen as a smear with a spectral distribution of colors across it, rather than sharply in natural colors.

This is shown in Figure 5.1. Bear in mind that until the last section of this chapter it will be assumed that the images are reconstructed with laser light of the same kind as was used in making the holograms, or with nearly monochromatic light of the same color derived by using a color filter with a white

¹We speak here of ordinary transmission holograms. Special types such as "rainbow" transmission holograms, which are white-light viewable, are discussed in later chapters of this book.



Figure 5.1. Here the same hologram as in Figure 1.1 is illuminated with a white light source. The smears seen in this black-and-white photo would appear as spectral color smears were the image viewed "live." (Photo by Dr. C. E. Hamilton, produced by Mr. Rainbows.)

light source. Under those conditions, the geometric model gives the right answers and is, in fact, very impressive in its explanatory power. The last section of this chapter will comment on the nature of the color problem and indicate how it will be dealt with in this book.

A GENERAL DESCRIPTION OF THE REAL IMAGE

In Figure 5.2 a diverged laser beam lights the hologram from the left. The hologram "processes" the light and *three* beams emerge. One beam diverges and travels upward. This beam produces the virtual image. Another beam converges, or focuses, and travels downward. This beam produces the real image. There is also an undeviated beam. This "direct" beam gets through the plate as a kind of leakage. It does not carry holographic information about the scene and therefore will not be discussed further.

If you look into the divergent beam you are able to see the virtual image because the lenses in your eyes provide the requisite focusing action. On the other hand, the convergent beam produces the real image without the need for any external lenses. A simple and interesting experiment consists of letting the beams fall on a white card or piece of paper. If the card is placed in the divergent beam, only a vague region of unfocused light will be seen. If the card

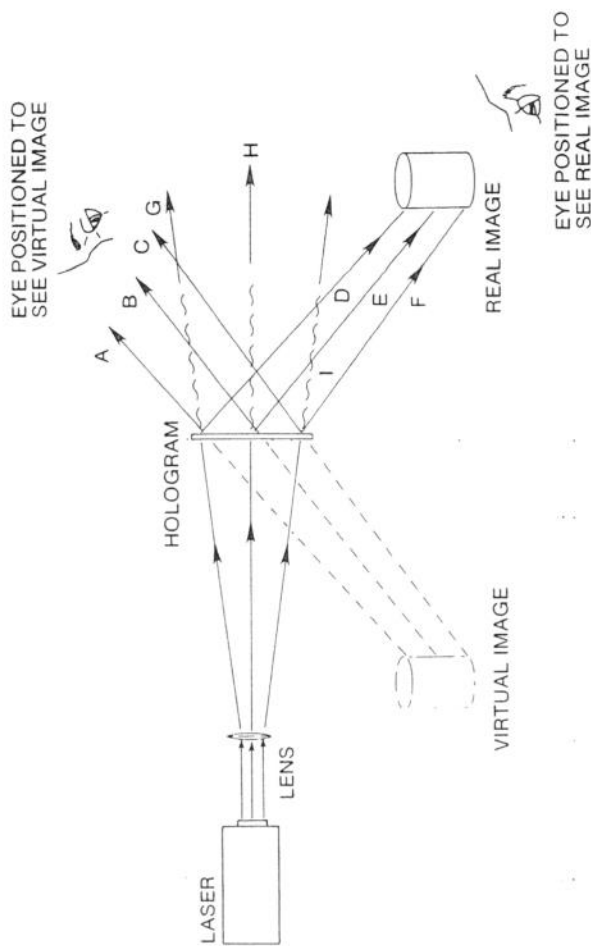


Figure 5.2. A transmission hologram producing two images. Rays A, B, C pertain to the virtual image. Rays D, E, F pertain to the real image. Rays G, H, I pertain to the direct beam.

is placed in the convergent beam, the objects in the scene *will* be imaged. This is the real image.

Since we are assuming that the readout light has the same color as that used to make the hologram, both images will be seen in that color. Ordinarily this means the images are in the characteristic red color of the helium-neon laser.

The real image has a number of special properties. One of these is that it exists over a considerable region of space. If you move the card closer to or farther from the hologram (perhaps even over a range of several feet), the image will still appear on the card. This means that the *depth of focus* is very large.

With the card in any position, the objects in the scene can be distinguished, but some will be more sharply in focus than others. As the card is moved, the in-focus parts change. Different parts are in focus at different distances from the plate but the card can show you only two-dimensional slices.

With the method of reconstructing the image shown in Figure 5.2, the objects in the image may be seen greatly magnified in comparison with the sizes of the original objects. For example, an object that was one inch tall may appear as one foot tall.

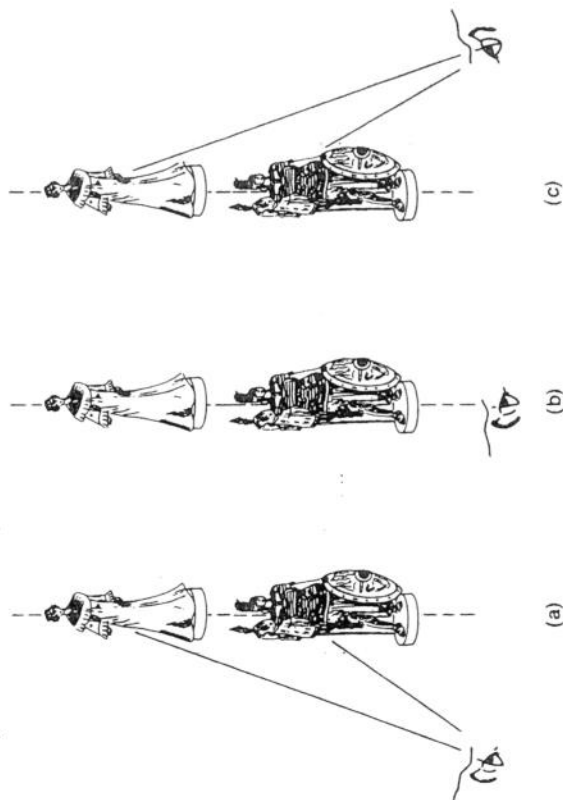
Finally, if you place your eye in position to see the real image with or

without the card as an intermediary, you see a very curious property of the image.² The effect is called *pseudoscopy*.

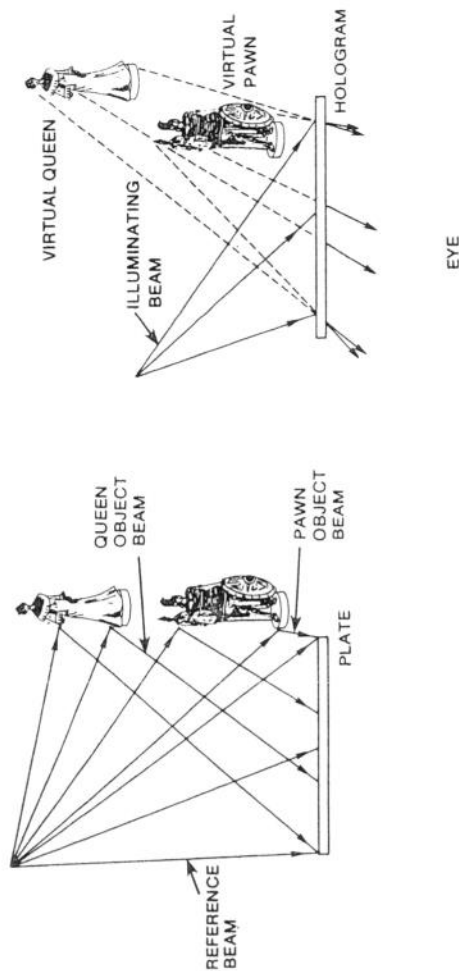
PSEUDOSCOPY IN THE REAL IMAGE

The pseudoscopic view is best understood by contrasting it with the more familiar *orthoscopic* view of real objects. In Figure 5.3 there are two chess pieces—a pawn and a queen. When your eye is placed to the left of the dotted line as in (a), it sees the pawn as closer to it than the queen, with the queen to the left of the pawn. If the eye moves to the right, the relative positions of the pieces will appear to change, with the queen appearing to move to the right of the pawn. You should be able to see this sequence shown in the figure and act it out by holding a finger of each hand before you and shifting your point of view.

Figure 5.3. (a) The viewer's eye sees the queen to the left of the pawn. (b) The eye has moved to the right and the queen is behind the pawn. (c) The eye has moved farther to the right, and the queen is now seen to the right of the pawn. In effect, from (a) and (c) the queen "moves" to the observer's right, relative to the pawn.

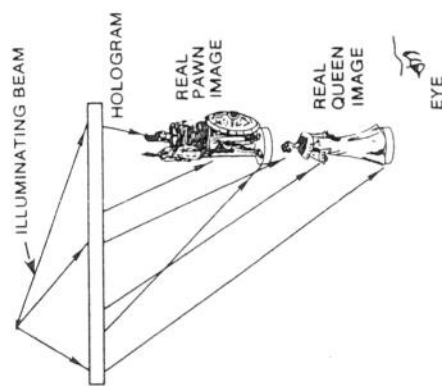


²It is best to use a card to locate the image before positioning the eye, otherwise considerably visual searching may be needed. With a low-power laser and a well-diverged beam there is no hazard in looking into the beam momentarily. The reader is urged to read Appendix I, "Laser Safety."



(a)

(b)



(c)

(d)

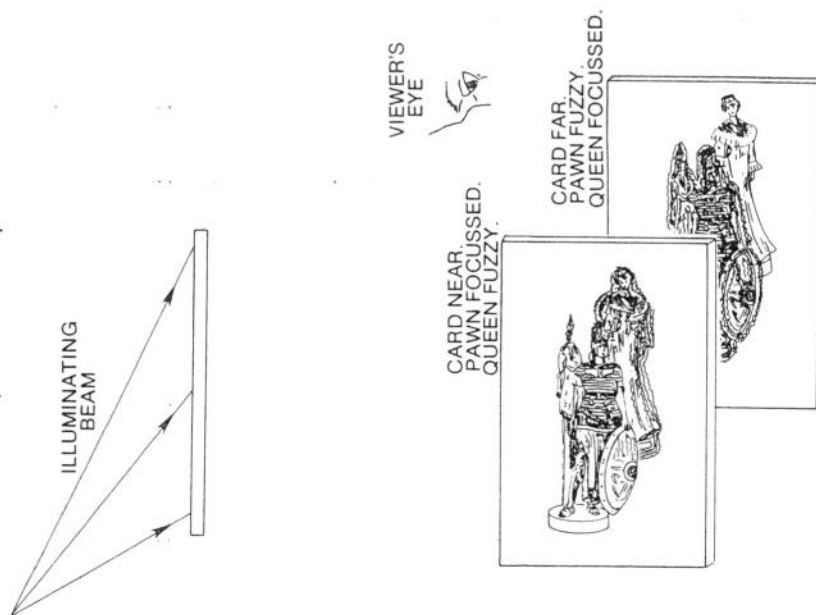
(e)

Figure 5.4. (a) Making a transmission hologram of a pawn and a queen. (b) Orthoscopic view of the virtual images. (c) Viewing the real images with the eye. (d) If the eye is near the plate, the pawn is in focus and seems nearer the viewer than does the queen. (e) If the eye is farther from the plate, the queen comes into focus, but the pawn still seems nearer to the viewer. Thus where the images really are and where they seem to be are opposite.

Notice that when the chess pieces and the eye are completely lined up as in (b), the pawn will block some of the body of the queen from view. There can be no doubt that the pawn is closer to the eye than is the queen. If the eye is shifted to the left, the queen also is seen to shift to the left relative to the pawn. These changes in the apparent left-right locations of objects are said to be *parallax* effects. This is the usual orthoscopic view of things.

Now let's consider the pseudoscopy in the real image. Figure 5.4 (page 53) shows again how a transmission hologram is made, this time using the pawn and queen as objects. In (b) the finished hologram is illuminated and the eye is placed to perceive the virtual image. What is seen is orthoscopic. The objects appear located as the real objects did in (a), and with normal parallax.

Figure 5.5. By using a card to locate the images, the pawn is in best focus nearer the plate than is the queen.



In part (c) the real image is being viewed. Remember that with the considerable depth of focus the chess pieces are not imaged as sharply localized as the drawing indicates, but rather there is a region over which the eye can be moved with the pawn and queen remaining discernible, though with fuzziness in one or the other. At one distance the pawn will be sharp, but superimposed as an obstacle on a relatively fuzzy queen. At another distance the pawn will be fuzzy and the queen sharp, yet the pawn will appear to be an obstacle between the observer and the queen. Figure 5.5 helps to clarify this. Here a card is used instead of direct viewing with the eye. As indicated, the evidence is that the pawn image is closer to the plate than is the queen image. However, if you move your eye sideways, you will see the queen (which you know by visual evidence to be behind the pawn) move oppositely from how you would expect according to ordinary orthoscopic experience. That is, if you originally see the queen to the left of the pawn and move your eye to the right, the queen will be seen to move to the left and not to the right. There has been a reversal of the parallax effect, in this sense.

Has there also been a reversal of the near-far relationship? If the eye looks at the virtual images of the pawn and queen in Figure 5.4 (b), the pawn would be perceived as nearer the eye than is the queen. In Figure 5.4 (c) you are looking at a real image of a pawn, with the image *farther* from your eye than is the image of the queen. Yet what you see gives the definite impression that the pawn image lies closer to you, as in parts (d) and (e). The pawn image is actually farther away, and yet it seems closer, in a sense. The near-far relationship is said not to be inverted, but because of the visual evidence, you could also say the opposite. This will be clarified in the next sections.

HOW THE REAL IMAGE IS PRODUCED

We have been describing in operational terms what is meant by pseudoscopy. It is time to explain how the real image is produced according to the geometric model, and how its peculiar nature can be accounted for.

Let's begin with the various parts of Figure 5.6. Part (a) shows a now-familiar drawing. The rays from source A reflect from surfaces in the hologram and diverge. If they are traced backward, they intersect at the original location of the object at B. In part (b) the directions of the rays in (a) have been reversed. It is a basic principle in optics that the rays in (b) will behave just as did those in (a), except that the light will travel in the opposite directions. It follows that if the plate is illuminated with the converging rays, they will reflect and then converge (focus). They form the real image at A.

This is the simplest way to account for the formation of the real image. Furthermore, it is only in this way that the most sharply defined and least distorted, or "aberrated," image is reconstructed. However, the real image can be formed if the hologram is illuminated with a laser beam *diverged* with a

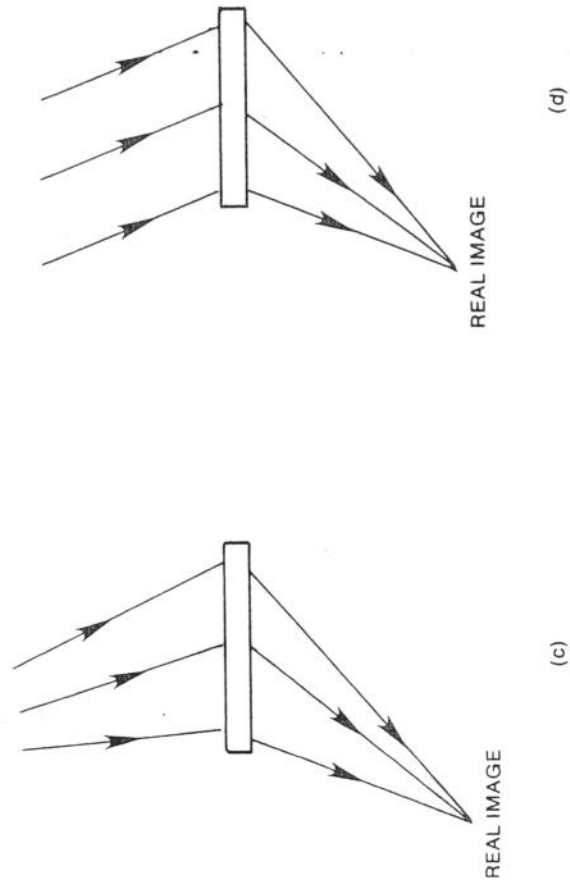
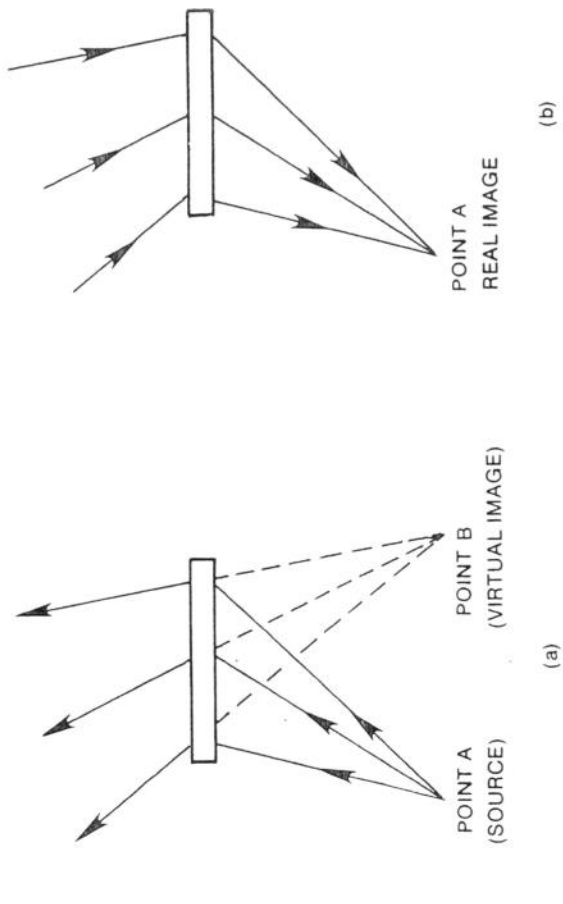


Figure 5.6. (a) Production of a virtual image of a pointlike object at B, using the same source A as was used in making the hologram. (b) The principle of reversal of light rays used to explain the formation of the real image. Instead of such convergent rays either divergent rays (c) or parallel rays (d) can be used.

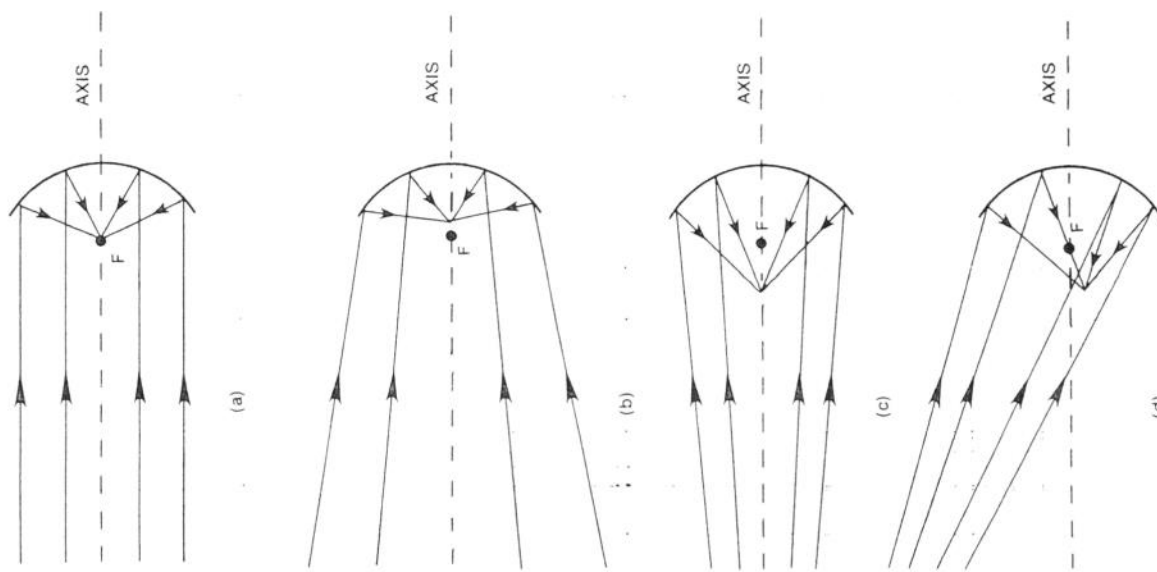


Figure 5.7. Focusing of various kinds of light beams by a concave mirror. (a) Parallel rays parallel to the axis of the mirror. (b) Convergent rays along the axis. (c) Divergent rays along the axis. (d) Convergent rays from an off-axis source.

simple lens, or even with a direct collimated laser beam. It takes great skill to arrange the illumination as described above, so a divergent beam is ordinarily used.

How can such diverse modes of illumination be used? Let's consider the following analogous case, which has the advantage that you can verify what is said if you have available a concave shaving mirror or the like.³ See Figure 5.7 (page 57).

If parallel rays along the axis strike the mirror as in (a) they come to a focus at point F, which is called the focal point. This can be observed by using the sun or a distant street lamp and a concave mirror. This works also if the rays are not parallel, as in (b) and (c). Even if the rays come from an off-axis source as in (d), an image will be formed. Generally these images will be aberrated or distorted in some way.

Similarly, the hyperbolic mirrors in a hologram can reconstruct the real image even with considerable variation in the nature of the illuminating beam. This is also true of the virtual image, as shown in Figure 5.1.

EXPLANATION OF PSEUDOSCOPY

Pseudoscopy in the real image is visual confusion about the near-far relationship and about the parallax effect. One of its surprising aspects is that if, as in Figure 5.4 (c), the pawn-image is farther from the eye than is the queen-image, the pawn seems to be nearer. Suppose that you were looking at real chess pieces, as in Figure 5.3. In order to see all of the queen from any one vantage point, light rays from all of the near side of the queen must reach the eye. Thus in part (b) of the figure, in making the hologram, some of the light from the queen is blocked by the opaque body of the pawn. This visual clue tells us that the pawn is between us and the queen. (The sensory apparatus also uses the fact that the eye's lens muscles must be used differently to focus on the objects.)

Consider Figure 5.4 (c) again. Now the eye is not looking at real objects, but rather is reacting to light rays coming from the hologram. Rays that converge to produce the pawn-image pass right through the region where the queen-image is located, and on to the eye.

However, when the hologram of the chess pieces was made, some of the light rays from the queen to the plate were blocked by the pawn. The hologram cannot contain any interference fringes to which contributions were lacking in the object beam. Therefore, when the real image is reconstructed, you see some blocking of the queen by the pawn. You naturally interpret what you see as indicating that the pawn is interposed between you and the queen.

You can now observe the parallax effect if you shift your eye sideways. You are looking at a pawn-image that is farther from you than the queen-image.

³Such a mirror is probably a spherical surface. The discussion refers to a mirror of a slightly different shape—a "parabolic" mirror, such as is used in reflecting telescopes.

You will see the pawn-image move with normal parallax effect but will be startled to find yourself thinking that the parallax effect has been reversed. Your surprise comes from the fact that you think of the queen as being farther away, rather than closer.

Although our example used two well-separated chess pieces, the same effects occur with a hologram of a single object. For example, a hologram of a plastic airplane model gives real images of the various points in the model, which are related as in the case of the chess pieces. If a strongly rounded object (such as the body of a flashlight) is used, you may get from the real image a sensation of looking at a concave object.

Michael Wenyon says that if you work with pseudoscopic images enough, you learn to convert what you see into orthoscopic images automatically.⁴ This is reminiscent of experiments in which people wore spectacles that made the world appear upside down to them. After an initial week of confusion, the people adjusted and saw the world as right side up despite the glasses.

DEPTH OF FOCUS

The meaning of *depth of focus*⁵ is well illustrated in the preceding discussions of the real image. A card placed anywhere in a range of distances from a hologram can pick up the real image. This is a case of large depth of focus. If the card had to be placed at one particular distance, the depth of focus would be small.

This subject is familiar to photographers. Even for nonphotographers the concept is easily understood if discussed in terms of a simple camera. Figure 5.8 shows representative light rays passing through a lens. In order to get a good photograph with this arrangement, the film can be placed anywhere within a certain region L without serious deterioration of the image. If the lens is wide open as in (a), the region L is narrow, but if a diaphragm is used to make the "entrance pupil" small as in (b), the region L will be wider. This "L" is the depth of focus. As indicated, it depends on the size of the lens opening. (It also depends on the ratio of the focal length to the diameter of the opening, a quantity known to photographers as the *f-number*.)

When light reflects from a curved mirror the reflected rays can converge to produce a real image. The process is not the same as that by which a lens acts on light passing through it to produce a real image, of course, but the two processes are sufficiently analogous for our purposes. You can get a useful conception of why the holographic real image has large depth of focus by

⁴Michael Wenyon, *Understanding Holography* (New York: Arco Publishing Company, Inc., 1978), pages 68-69.

⁵*Depth of focus* is not the same as *depth of scene*. The latter has to do with how deep a scene made up of real objects can be to make a hologram of it. Depth of focus is also known as *depth of field*.

Notice that the question of depth of focus does not arise in connection with the virtual image. In viewing this image, your eye receives a divergent beam of light and focuses to give you a sharp view, as it would if it were receiving light from an actual set of real objects. In the reconstructed scene you may pay for focusing closely on parts of the scene by losing some sharpness in the clarity with which other parts are seen, but this is true also in looking at real objects.

MAGNIFICATION

The real and virtual images of an object do not necessarily have the same size as did the original object. It is sometimes said that the virtual image perfectly reproduces objects with their proper sizes. In fact, this is not always so as can be seen by a simple experiment—if you shine diverged laser light on a hologram and then change the distance from the laser to the hologram. Changes in the dimensions of objects in the scene will be seen, and not all of these dimensions can be correct.

The *magnification* is the ratio of the image size of an object to the actual size of the object itself. Thus magnifications greater than or less than one mean, respectively, that the image is larger or smaller than the original object. Only if the magnification is equal to one will the image show the objects in their true sizes.

There are several variables that affect the magnification: the distances of the objects and of the source of light from the plate during the making of the hologram; the distance of the source of light from the plate during reconstruction of the scene; and the wavelengths of the light used in making the hologram and of the light used in viewing.

There are two special cases in which the magnification will be equal to one. In one of the cases the light source used in making the hologram and the light source used in viewing the scene have infinite distances from the plate. That is to say, plane waves are used in both cases. This condition is not commonly met because laser light diverged by a lens is usually used.

The second special case occurs when the recording and reconstructing sources have the same finite distances from the plate. In other words, the wavefronts are spherical and alike. Obviously if you set up a hologram made with an arrangement about which you knew nothing, it is unlikely that if you illuminate it at your convenience you will duplicate the recording conditions. In the little experiment described above you can expect to see the virtual image change in size by a factor of two, three, four, or so. If the two distances are not equal, the magnification for the virtual image will always be less than one. For the real image, the magnification will be greater than one.

Up to this point, we have assumed that the wavelength of the reconstructing light and the wavelength of the viewing light are the same. This is

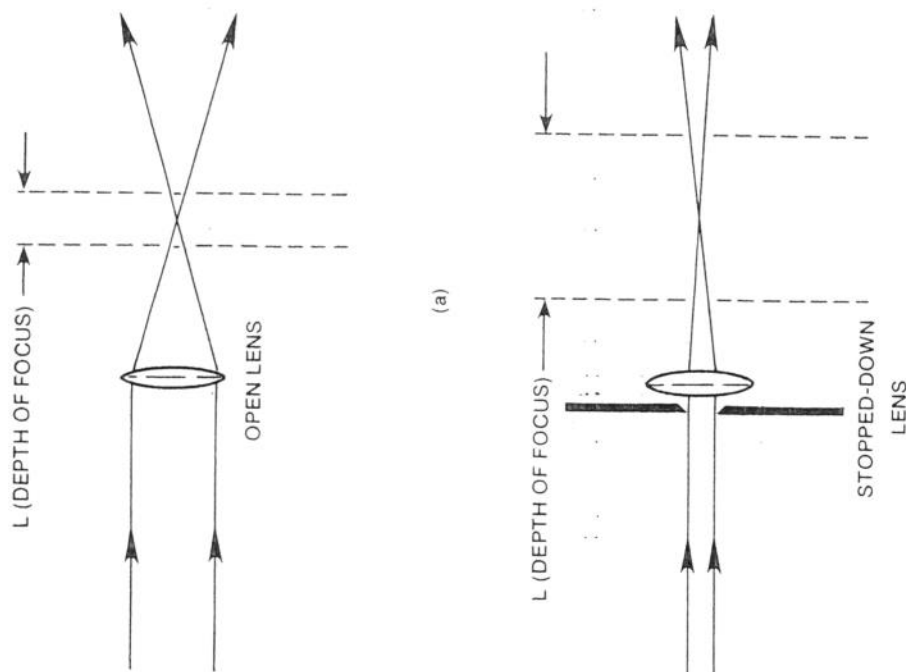


Figure 5.8. (a) A lens at full aperture has relatively small depth of focus. (b) When the same lens is stopped down it has larger depth of focus.

thinking of the reflecting surfaces in the hologram as having large focal lengths (since they are nearly flat) and as being much "stopped-down" (since the emulsion is so thin that it contains only a small width of a surface across it). A photographer would say they have very large f-numbers.

always the case when a helium-neon gas laser is used in both instances. If different wavelengths are used, the magnification depends on the ratio of the wavelength used in viewing to the wavelength used in making the hologram, in addition to the source distances.

In principle this makes possible the achievement of large magnification. For example, if a coherent source of short wavelength X-rays were available and used in making holograms, and visible light used to view them, magnifications of the order of one million would be possible. Such sources do not exist at present so this scheme has not been brought to reality.⁶ When visible light of the most extreme wavelengths possible are used, a magnification only of the order of two can be achieved in this way.

An effect that accompanies change in size of images is that they are nearer or farther from the viewer, according to the magnification. With large changes in the magnification, this can be objectionable. An example is the case of acoustic holograms, which are discussed in Chapter 10. With these the wavelengths used in making the holograms and in the viewing of them are enormously different and the final images are seen as though they were very remote.

REDUNDANCY: EACH PIECE OF A HOLOGRAM IS A HOLOGRAM

Consider Figure 5.9. The wide and narrow holograms in (a) and (b) both capture and record interference fringes from the object and each contains a holographic record of the scene. However, in (a) the plate "sees" the object from a wider

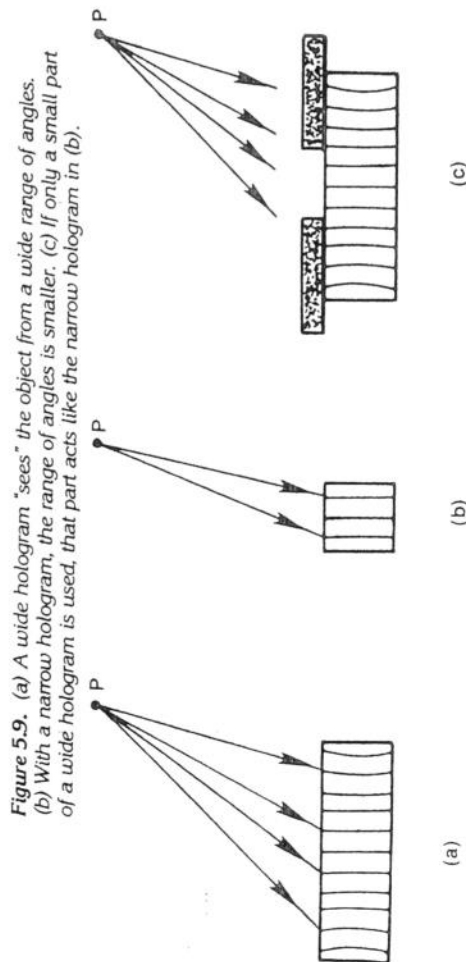


Figure 5.9. (a) A wide hologram "sees" the object from a wide range of angles. (b) With a narrow hologram, the range of angles is smaller. (c) If only a small part of a wide hologram is used, that part acts like the narrow hologram in (b).

(a)

(b)

(c)

⁶"Pictures" of individual atoms have been made holographically using X-rays, but by a special method. For more about this, see Chapter 11.

range of directions than in (b). On reconstruction of the virtual (or real) image with the wide hologram, the viewer can look at P from a range of directions (as though looking through a wide window). With the narrow hologram, the object at P will be seen three-dimensionally again, but you, the viewer, will have little freedom in moving your eye to see it.

If a hologram is broken into pieces, each piece will reconstruct the scene. *Redundancy* is the term applied to this. A less destructive way to observe this is to use the usual diverged beam and an opaque screen with a hole in it to block all but a small part of the plate. Still another way is to let the undiverged laser beam (no lens) fall on the plate. Only a small spot will be illuminated and with some effort you can manage to peer through this to see the virtual image. By using a card, you can discern the real image. (In either case, be sure that you are really looking at an image due to the actual laser beam. Accidental reflections from parts of the beam exit port of the laser may be bright enough to illuminate larger regions of the plate and give images.)

Besides the limitation in the range of viewpoints possible with small pieces of a hologram, there is also some degradation of the quality of the image. (Obviously the inherent graininess of the emulsion becomes significant with very small pieces.) The total number of fringes in a hologram, and not just the spatial density of fringes, is important in determining the image quality. Big is generally better than small in this context.

THE HOLOGRAM IS TWO-FACED

Suppose a transmission hologram is set up with a laser beam shining on it. If you turn the plate front-to-back so the side that was facing the laser becomes the side you look through, you will again be able to see the virtual and real images.

Figure 4.1 provides a basis for a brief discussion of this effect. Suppose that besides the plate shown there were another one displaced vertically downward to occupy a position symmetrically located with respect to an imaginary line between A and B. The interference pattern recorded would be like that in the plate shown, but related to the faces of the plate in front-to-back fashion. If the plate that is shown (after processing) were flipped over without moving it otherwise, it would act as a hologram as would the other plate in its position.

The case illustrated is a highly symmetrical one and so is rather special, but this discussion of it should convey the basic idea.

DEPTH OF SCENE AND COHERENCE LENGTH

Most transmission holograms provide views of objects that are small and compactly grouped together. They have small *depth of scene*. This is not due to a limitation in the fundamental nature of holography but rather is associated with

a certain property of the laser beam. A laser beam (or any other source of light for that matter) has a limited *coherence length*. This limits the depth of scene that can be achieved.

Refer to Figure 5.10 for an explanation of these matters. The waves *a* and *b* are identical with respect to wavelength and amplitude, and they are in step along their lengths. At such places as A, B, and C they would interfere constructively if superimposed. The result would be the wave *c*, which has the same wavelength but greater amplitude or intensity. Now consider the waves *a* and *b* of Figure 5.11. The wave *b* has a slightly shorter wavelength than the wave *a*. At A, the waves both have crests, and they would interfere constructively there. In the region B the waves have fallen somewhat out of step, and at C they are still more out of step. Eventually at X the waves are completely out of step. The condition for constructive interference at A becomes that for destructive interference at X. Although the drawing does not show it, after traveling another distance equal to that from A to X, the waves will be back in step again. The distance from in-step back to in-step again is twice the distance from A to X and is called the *coherence length*.

Any light source that is not perfectly monochromatic has a limited coherence length because there are constituents in its light that fall out of step along the beam. This is true of the popular helium-neon gas laser, which actually emits more than one wavelength. The wavelengths do not differ greatly but their interplay limits the coherence length. The numerical value of the coherence length depends on the length of the tube in the laser and so is not the same for all helium-neon lasers. Typically, it is several tens of centimeters in magnitude.

Figure 5.10. The waves *a* and *b* are identical and in step along their lengths. They interfere constructively everywhere and their resultant is the wave *c*.

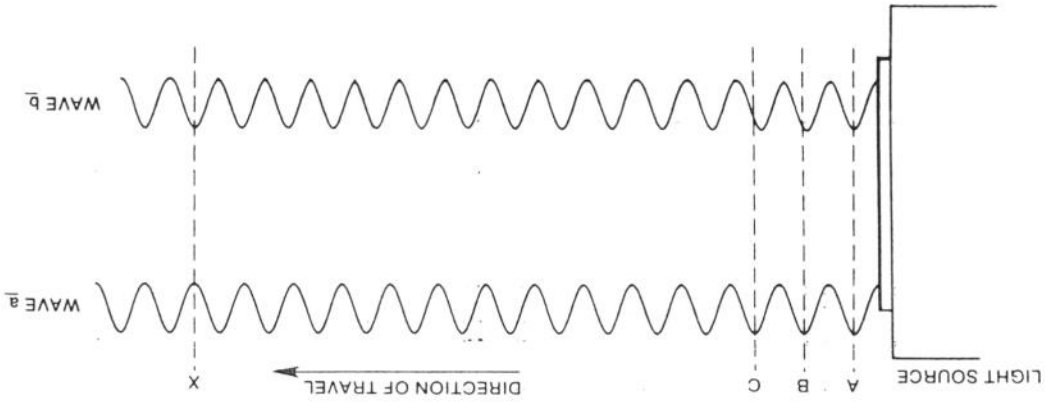
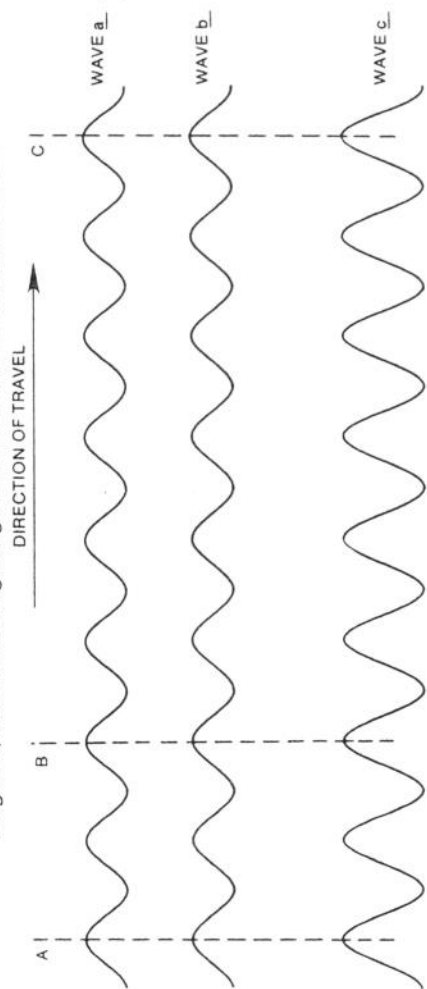
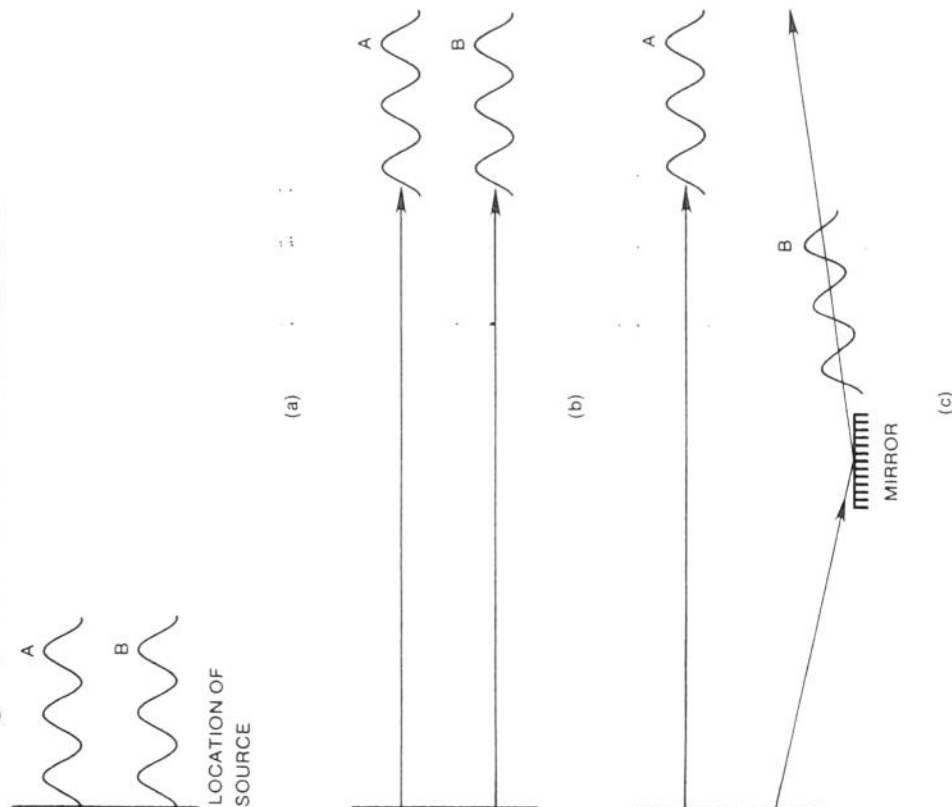


Figure 5.11. Waves *a* and *b* differ in wavelength. At A they are in step. Then they become more and more out of step, as at B and C. At X they have become completely out of step.

(For helium-neon lasers with various lengths, the coherence length decreases as the length increases.)

Special lasers incorporate optical devices that suppress some of the wavelengths and can have much greater coherence lengths, possibly extending to many meters. Lasers with coherence lengths of as much as a kilometer (about

Figure 5.12. (a) Two burst of light are emitted in phase with each other. (b) The bursts have traveled equal distances in the same time and are still in phase. (c) A mirror is used to make the paths unequal, and in a given time interval the burst B lags so far behind A that the bursts cannot interfere at all.



three-fifths of a mile) exist. Such special lasers are much more expensive than ordinary helium-neon lasers.

In making a hologram the size of the scene is determined by the coherence length of the laser. For example, if the coherence length is one-half of a meter, the depth of the scene must be substantially less than that, lest parts of the scene be not holographed. Thus the practical holographer should know what the coherence length of the laser is. A simple method is given in Chapter 14.

The rest of this section deals with the physical basis for the relationship between coherence length and the making of holograms.

To begin an explanation of this let's refer to Figure 5.12. In (a) two identical short wavetrains start out in phase. Each will be referred to as a "burst." In part (b) the bursts have traveled equal distances to the right in the same interval of time, and they are shown as side by side. If they were superimposed, they would interfere. In part (c) of the figure the case is quite different. The burst A has traveled a certain distance and the burst B has traveled the same distance in the same time, but thanks to the different geometries of the paths, B has not come into position to meet A. In fact, B lags so far behind A that there is no overlap of the two bursts at all, and no interference can take place.

After B travels a sufficient distance further, it will arrive where A was, but the interim A will have moved on. Whether A and B overlap and can interfere at a point depends upon whether A and B travel paths of the same length from their mutual source to the point in question. If the paths are slightly unequal, there will be partial overlap and weak interference. If the paths are sufficiently unequal, there may be no overlap at all.

The next step is to take the bursts to correspond to samples of light from a laser (or any other source) with limited coherence light. This puts us in a position to understand Figures 5.13 and 5.14. In the first of these the object beams O_1 and O_2 take paths of different lengths to get from the lens to the holographic plate than does the reference beam R. Depending on what the path differences are, you might get no hologram at all, or one with very low contrast. In Figure 5.14 the configuration has been corrected by using a front surface mirror to lengthen the path of ray R in the reference beam from the lens to the plate. In addition, the objects being holographed have been moved closer together. (The screen is present to block laser light from reaching the plate directly, rather than after reflection from the mirror.)

The golden rule is that the maximum path length difference for the reference and object beams should be made as small as possible.⁷ Equalization cannot be brought about for *all* rays, as you can see by considering the various points such as P_1 and P_2 on the plate, and the necessary spread of points in the scene. Hence the rule refers to the *maximum* path difference.

If the maximum path difference is not zero but is an integral number of

⁷A mere piece of string becomes a valuable holographic tool by providing a simple way to compare path lengths.

coherence lengths, the condition is suitable for overlap of the "bursts." However, such a setup is of more theoretical than practical interest.

Holograms of very deep scenes have been made, such as of all the things in a room. If you see one, however, you can be sure it was made with a special laser. The coherence length must have been great and the laser probably cost tens of thousands of dollars.

At the other extreme are ordinary light sources. These have some coherence length but may amount to only a fraction of a millimeter. Even if a color filter is used to make the light more nearly monochromatic, the coherence length will still be very short. In this connection it is interesting to know that when Dr. Gabor made the world's first hologram, long before the advent of the laser, he used a mercury lamp with a color filter, but even then had to use a very flat object. (It was a piece of thin transparent material with dark lettering on it.)

MULTIPLE-SCENE HOLOGRAMS

In ordinary photography a double-exposure is usually due to an error and is to be avoided. It results in a confusing superposition of pictures. In contrast, multiple exposure of a plate in holography is often done deliberately. One application of the technique is in an important specialized branch of holography called interferometry or holometry. This subject is discussed in Chapter 11. This section discusses another—and potentially exceedingly important—application known as *multiple-scene* or *multichannel* holography.

Consider the simplest case in which two scenes are recorded in a single transmission hologram. To observe one of the scenes, the hologram is oriented suitably in a laser beam or in color-filtered light from a white light source. To observe the other scene, the hologram is reoriented. In each case only one scene is visible, with no "cross-talk" from the other. For example, in one commercially available two-channel hologram, a model of a motorcycle is seen with one orientation, and a model of a tank after a rotation of the plate.

How such holograms are made and work is easy to understand. With one of the scenes set up, a transmission hologram is made but the emulsion is not processed. While the emulsion is protected from exposure to light, the scene is changed and the plate is reinstalled with a rotation in the setup. The plate is exposed again. The plate is then processed to make the final hologram.

The result is that there are two distinct sets of interference fringes in the hologram. These act on a readout beam of light so differently that there is no overlap in the images. Each image can be seen independently of the other. That this works is evidence of a great difference between holography and ordinary photography.

If two-channel transmission holograms can be made, then how about multichannel recording of greater complexity? How about 500 images in a

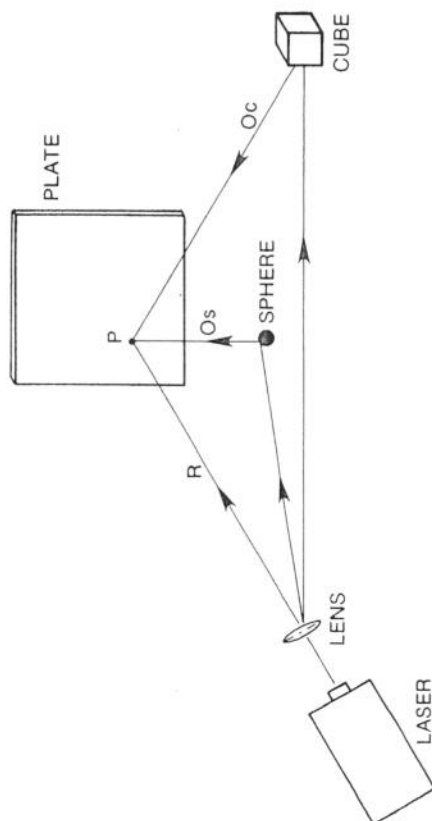
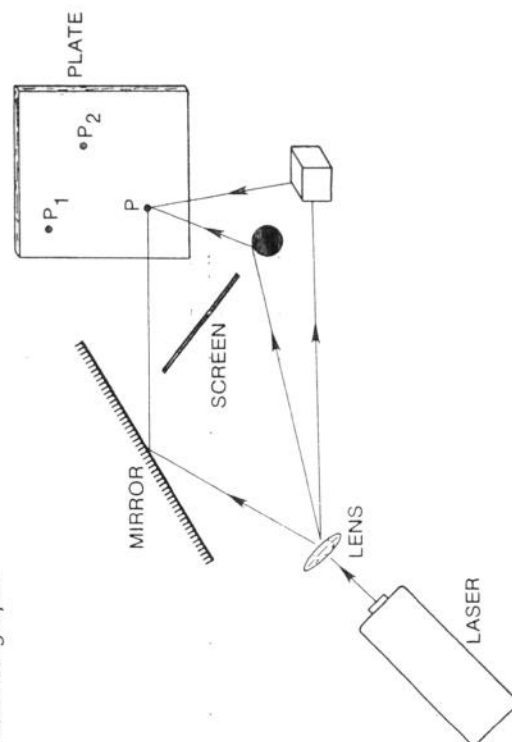


Figure 5.13. The reference beam is represented by ray R. The object beams from the sphere and the cube are represented by waves O_s and O_c respectively. The three rays travel different total distances from the lens to the point P on the plate.

Figure 5.14. The reference beam path length is increased by the front surface mirror, and the sphere and cube have been moved to make all three path lengths more nearly equal.





Reflection holograms

INTRODUCTION

In viewing the virtual image with a transmission hologram, you look through the plate as through a window, with the light source on the other side. In viewing a reflection hologram, the light source is on your side of the plate, and the situation is more like looking at a mirror. The geometric model alone does not prepare you to understand some striking differences between the two cases. The differences exist in part because an optical effect called *Bragg reflection*¹ is not taken into account by the purely reflective geometric model, though it predominates with reflection holograms and is present to some extent in transmission holograms. Our purpose here is to explain this optical effect and then to apply it to reflection holograms.

SOME GEOMETRICAL OPTICS AND REFLECTION HOLOGRAMS

Let us start by thinking about how the purely geometrical model (aside from Bragg effects) explains the formation of images by a reflection hologram. (See Figure 4.13.) The light reflected from the surfaces in the hologram is perceived by the viewer as if it came from the virtual image. The reflecting surfaces in a reflection hologram are oriented along the plate, and this is quite different from

¹This effect is usually called *Bragg diffraction*, but in the present context our name for it is more descriptive of the phenomenon we are discussing.

the across-the-plate orientation of transmission holograms. However, the basic idea of reflections to produce the virtual images applies in both cases.

The production of real images by transmission holograms and by reflection holograms are quite different. A transmission hologram illuminated as in Figure 5.1 gives a virtual image and a real image simultaneously, which is understandable according to the geometric model. There is also a direct beam straight through the plate from the light source that carries no image information. What happens with a reflection hologram is shown in Figure 6.1. In (a) the viewer holds the hologram in a readout beam of light to see the virtual image. In this particular arrangement there is no real image.

However, there is an arrangement which gives a real image. To see it, the reflection hologram must be flipped over so that the faces A and B are interchanged, as shown in (b). This real image is pseudoscopic, while the virtual image seen as in (a) is orthoscopic.

Only the effects are described here; their explanations are given later. But if you have on-hand a suitable reflection hologram, you can easily make observations that are pertinent. The hologram must be viewable from both sides without a permanently attached opaque backing. By viewing it from one side and then the other you can see the images as in (a) and (b) in Figure 6.1, and their orthoscopic and pseudoscopic natures. This is more striking with some holograms than with others, and it is best if the object has substantial depth. We have shown the effect to many people, using holograms of statuettes, ancient coins, and so on, and have found that some people react strongly to the curious contrast between the two images. Evidently some degree of sophistication is required for the reaction to occur. In any case, it is something that any student of holography should experience.

AN OVERVIEW OF THE NATURE OF BRAGG REFLECTION

In an actual reflection hologram, the partially reflecting surfaces will form a complex array in the plate, even for a relatively simple holographed object. As you start to consider the optical effects that take place in reflection holograms, it is best to think of an idealized case in which the surfaces are plane, parallel, and uniformly spaced. Figure 6.2 shows such a stack of surfaces, with a beam of monochromatic coherent light falling obliquely on the top of the stack. When the light encounters the top surface, some of it is reflected back upward and to the right, and some passes through to the next surface. At this second surface some more of the light is reflected and some is transmitted toward the third surface. Similar reflections and transmissions occur at subsequent surfaces, though with decreasing amplitudes due to losses in the light.²

²For simplicity, we ignore multiple reflections of light back and forth in the region between any two adjacent surfaces and effects of refraction.

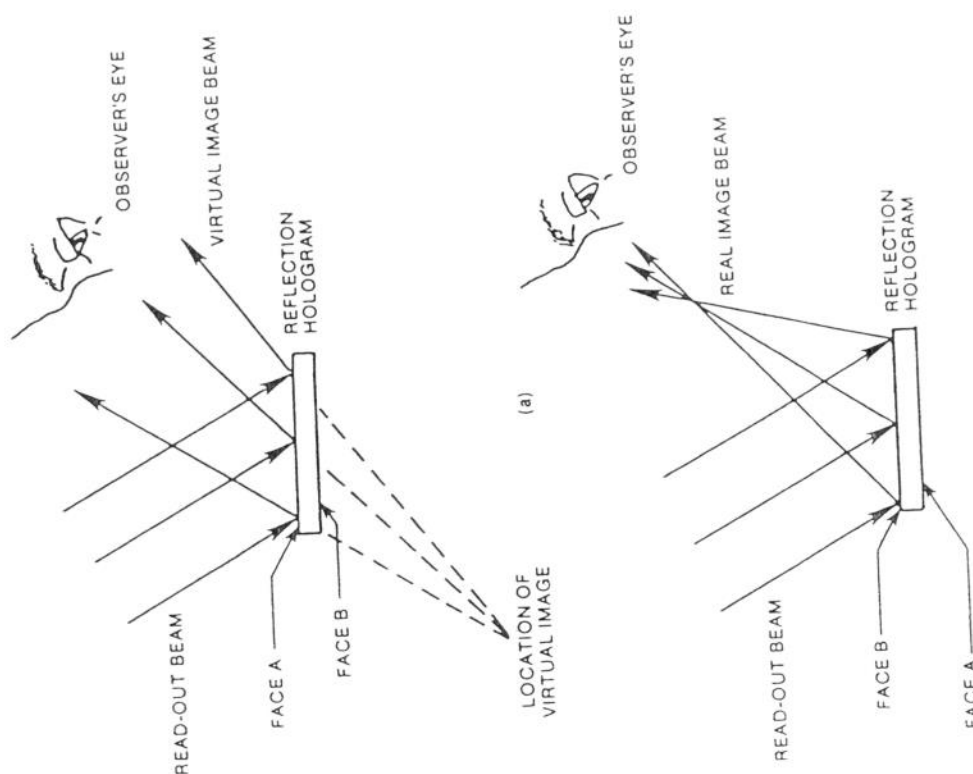


Figure 6.1. (a) When one side of the reflection hologram is used, the image is virtual and orthoscopic. (b) When the other side is used, the image is real and pseudoscopic.

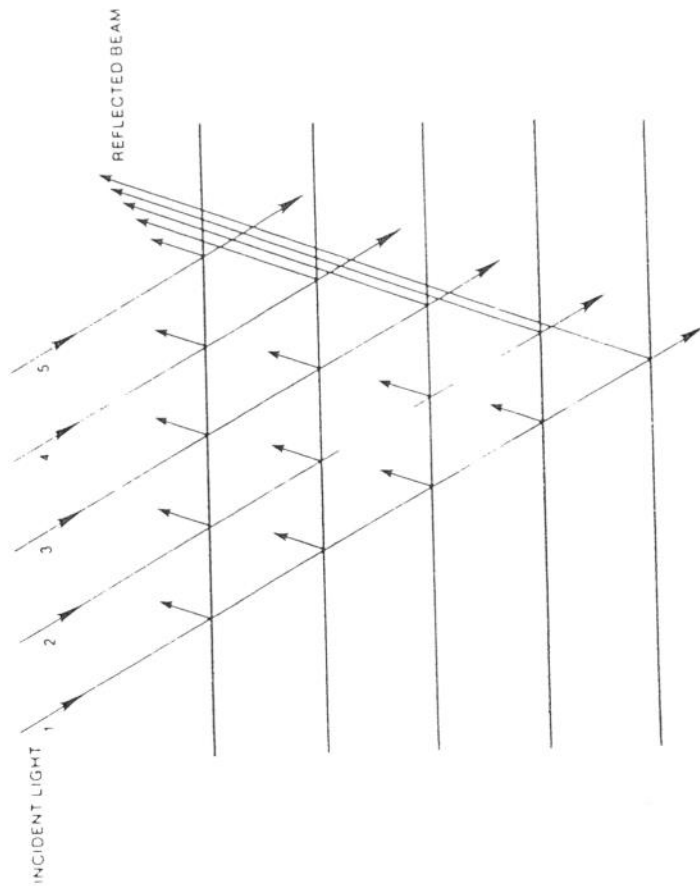


Figure 6.2. The reflected beam from the stacked surfaces is the result of light reflected from the various surfaces and interfering with each other as they emerge from the stack.

By now you know that if two coherent light beams are superimposed, there will be interference effects. If many coherent beams are superimposed, there will also be interference. You can correctly anticipate that the interference of the reflected beams will be important. This combination of multiple-surface reflections and the consequent interference is Bragg reflection.

BRAGG REFLECTION IN SOME SPECIAL CASES

Let's take up first the case of just two partially reflecting planes with the light falling perpendicularly on the top surface as in Figure 6.3. In (a) some of the incident ray *a* reflects directly as ray *b*, and some goes on as ray *c*. In (b) of

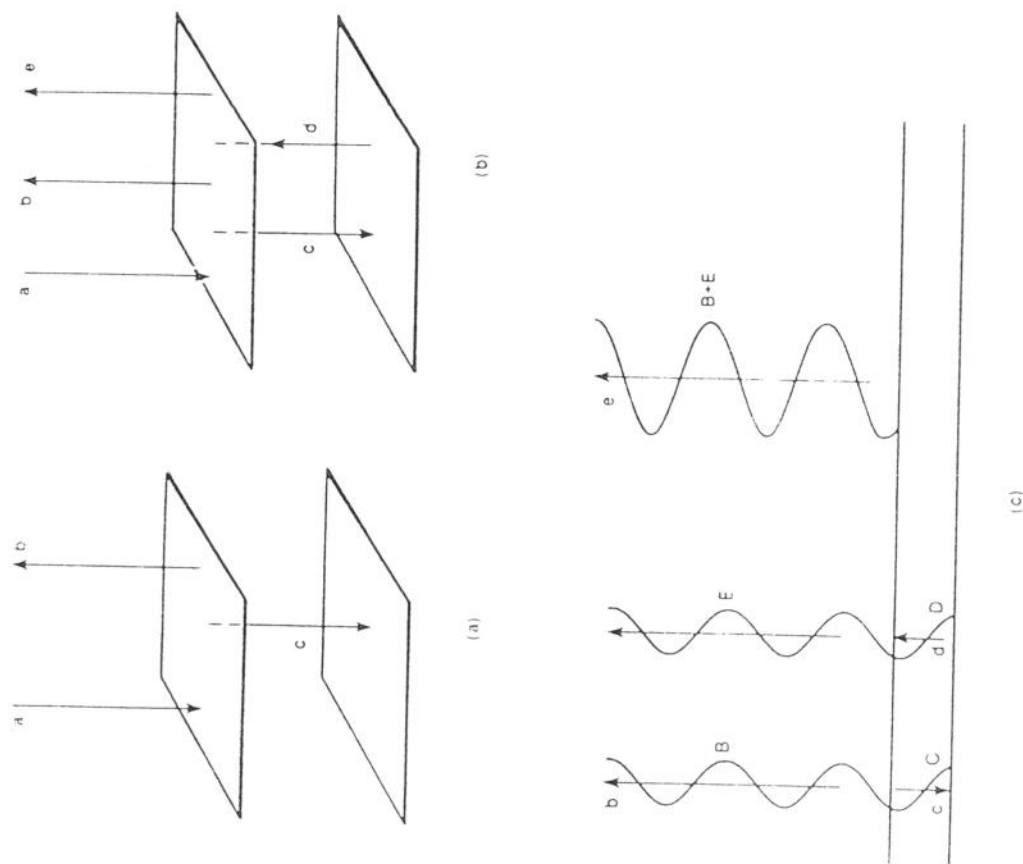


Figure 6.3. With normal (perpendicular) incidence on two parallel surfaces, constructive interference in the reflected light takes place when the wavelength is twice as great as the distance between the surfaces. For interpretation of the parts (a), (b), and (c) of the figure, see the accompanying text.

the figure, d is that part of ray c which reflects upward, and e represents this light after it emerges from the upper surface. With an extended beam of light falling on the upper surface, there will be many rays like b and e traveling upward, and superimpositions will occur. For simplicity in the drawing only b and e are shown, and these are shown as displaced sideways. You need to think of them as superimposed.

Now turn to a wave description of what is happening. In (c) of the figure the wave B corresponds to the reflected ray b , and the wave C to the ray c . Wave B is traveling upward, and C is traveling downward. Wave D corresponds to ray d and is traveling upward after reflection at the second surface. Wave E is the same as D after emergence. It is the upward waves B and E that interfere. The result of the interference is shown as the wave $B + E$. This is an upward traveling wave with greater amplitude than either B or E . In order for this to happen (constructive interference), a condition is that one-half wavelength of the light fits precisely into the space that separates the surfaces. This special case has been used in making the drawing. In this special case the wave B arrives to emerge from the upper surface in step with the reflected wave E .

What we have described is a strongly reflecting layer. On the other hand, if the wavelength of the light is made twice as great as before, each of the wave segments that correspond to C and D would fit with only $1/4$ of their wave cycles into the depth of the layer. When the waves E and B are superimposed (traveling upward above the surface), they will be $1/2$ wavelength out of step. The condition for destructive interference will be met. Now we have a non-reflective layer.

If light of various wavelengths falls on the layer, some colors will be strongly reflected, and some will be suppressed in the reflected beam. When the wavelength is such that an integral number of wave cycles fits into the total distance traveled down and then back up through the layer, there will be constructive interference. When the wavelength is such that an odd-half integral ($1/2$, $3/2$, $5/2$, ...) number of cycles fits in, there will be destructive interference. For intermediate wavelengths, there will be intermediate degrees of interference.

With this generalization made, you can understand that the layer will be a color selector or interference filter, for if white light shines on the layer, there will be color-preferential reflection. Such an effect is not rare in daily life. The characteristic rainbow colors seen with soap bubbles and with oil slicks are due to a similar phenomenon.

Now let's consider the case in which there is a stack of partially reflecting surfaces, but still with light falling perpendicularly on the top surface. This is illustrated in Figure 6.4. The wavelength is such that $1/2$ wavelength fits into each of the intersurface distances d . The waves W_1 through W_5 are those that emerge traveling upward after reflections in the stack. As they emerge, their various amplitudes are smaller and smaller, according to their depths of penetration and consequent losses. But they are in step, and as a result there is

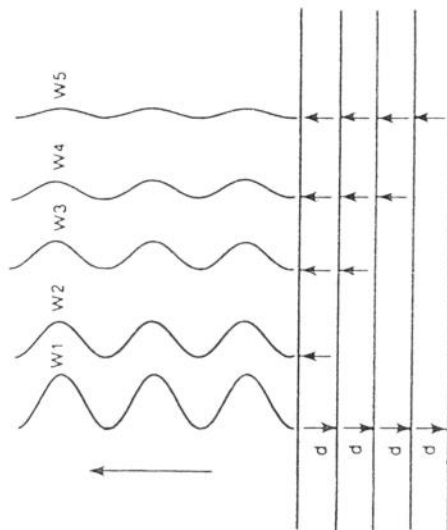


Figure 6.4. The waves W_1 through W_5 emerge from the stack in step with each other after reflections at various depths, if the wavelength is two times the distance between adjacent surfaces

constructive interference. Because the interference involves a number of cooperating beams of light, the reflection will be strong.

In the opposite case of destructive interference—for some other wavelength, of course—there will be very effective mutual cancellations. The multiple-layer system will be a highly selective color filter.

OBLIQUE INCIDENCE OF LIGHT

Figure 6.5 shows an extended beam of monochromatic light falling obliquely on stacked surfaces. (For simplicity only two surfaces are shown.) A and A' are two rays in the incident beam. Some of A reflects immediately as B , and some of A' as B' . Some of A passes through the upper surface, travels on as C , reflects as D from the lower surface, and finally emerges as E . B' and E are superimposed as they leave the stack traveling upward, and they will interfere.

Whether the rays B' and E interfere constructively, destructively, or in some intermediate way, depends as before on the wavelength and on the intersurface distance, but now it also depends on the angle of incidence of the light. For some angle of incidence there may be strong reflection, with B' and E in step under the given conditions. For some other angle of incidence, with all the other conditions remaining the same, there may be no reflection, and

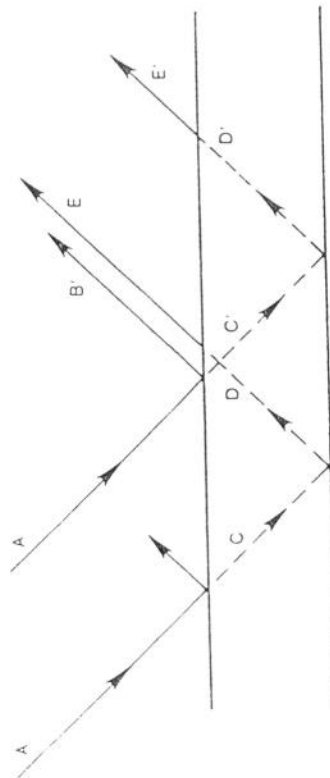


Figure 6.5. With oblique incidence of the light, the rays B' and E' (reflected from the first and second surfaces, respectively) interfere in a manner that depends on the wavelength, on the surface separation, and on the angle of incidence.

B' and E' may be completely out of step. If B' and E' are in some intermediate phase relationship between the two extremes, there will be some intermediate intensity of reflection.

If white light is used, there will be different angles for maximum intensity of the reflected light for the various color components in the illuminating beam. For example, the conditions shown in Figure 6.4 may be right for strong reflection of light of one particular spectral color, which the Bragg effect selects from the full range of colors in the incident light. If the angles are changed, the Bragg effect will select another color to predominate in the reflected beam.

This is the manner in which color effects seen with soap bubbles and oil slicks can be explained. In an introductory physics textbook such things are explored in detail. We will not have to do that here because holograms do not meet the usual textbook-case conditions. In holograms the stacked surfaces are not flat, parallel, uniformly spaced, all equally reflecting, and in fact are so granular on a microscopic scale that they are not even continuous surfaces.

THE BRAGG EFFECT WITH REFLECTION HOLOGRAMS

Figure 6.6 shows a model of a reflection hologram. Sheets of ordinary 8½-by-11-inch paper are spaced vertically with a separation of about 1/2 inch between adjacent sheets. These represent the interference fringes across the depth of a typical holographic emulsion. Among the most commonly used emulsions are Kodak SO-173 and Agfa-Gevaert 8E56 and 8E57. These are about 6 millionths of a meter thick. The separation between fringes is about 1/2 wavelength in typical cases. For the usual red laser light this separation is about 1/4 millionth of a meter. Hence the model is fairly realistic vertically.

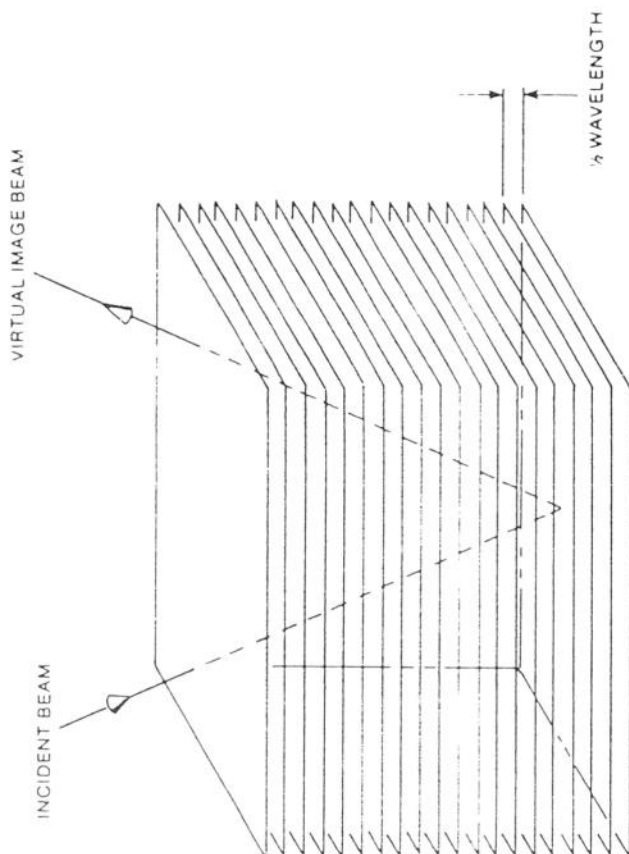


Figure 6.6. The paper sheet model of a typical reflection hologram.

With these dimensions in mind, let us consider what happens when light falls on a hologram of this kind. In Figure 6.6 only one ray is shown—the one that penetrates the full thickness of the plate before reflecting. In the full situation, light is partially reflected and partially transmitted from each of the surfaces, and the large number of surfaces favors strong Bragg reflection.

Suppose the hologram was formed and illuminated with red laser light. In the idealized case we discussed earlier strong Bragg reflection will occur for one precise angle. With an actual hologram it takes place over a wide range of angles. In a typical case, you may be able to see the image even through a range of angles as large as 20 or 30 degrees.

This difference between what really happens and what you expect on the basis of the simple model is attributed to the combined effects of the thickness of the emulsion, the density of the surfaces, the reflectivities of the surfaces, and the nonlinear nature of the recording medium.³ The theory shows that with

³A more detailed (but mathematical) discussion of this can be found in *Optical Information Processing and Holography* by W. Thomas Cathey (New York: John Wiley & Sons, 1974) pages 150-175.

an actual hologram, if the angle is that angle Λ_0 predicted in the idealized case, then the reflected light is most intense. If the angle differs from that ideal-case angle Λ_0 , there will still be Bragg reflection, but with the intensity of the reflected beam dropping off as the deviation of the actual angle from Λ_0 increases.

It is interesting that a hologram made with red light can be viewed with monochromatic light of a different color. To do this, it is necessary to tilt the plate in the beam suitably, since the Bragg angle is different for different wavelengths. Few people have access to lasers that emit light of various colors, but the strong yellow light from sodium lamps can be used to view a hologram in a color other than red. Color filters and ordinary light sources tend to give such low initial intensity that the color-angle effect may be hard to see with them.

We have just spoken of the use of nonlaser sources for viewing reflection holograms, although in our discussion of the theory of Bragg reflection, we had to assume the light to be completely coherent. It is indeed the case that reflection holograms can be viewed without a laser, and furthermore, ordinary white light can be used. This is a great advantage. Whereas a transmission hologram requires laser light, or at least highly monochromatic light, a reflection hologram can be easily viewed with sunlight or light from a small incandescent lamp.

Reasonably monochromatic light has some temporal coherence. In fact, the coherence length can considerably exceed the thickness of the emulsion. If the source of light is small in size, there will also be adequate spatial coherence. Bragg reflection can take place. But even white light has sufficient coherence. Given the small spacing of the surfaces (only about $1/4$ millionth's of a meter) and the limited thickness of the emulsion (about 6 millionth's of a meter) a coherence length even as small as $1/10$ millimeter is large!

Thus reflection holograms are white-light viewable, which makes them easy to view and exhibit in holography shows, and which makes feasible their use as desk ornaments, pendants, etc.

COLOR EFFECTS

"White-light viewable" does not mean "viewable in full color."⁴ If a hologram was made with red laser light and then illuminated with white light, the image would be seen in red. The hologram acts like a red-selecting color filter. The red image can be seen over a range of angles that may deviate considerably from that predicted by the simple theory.

Actually, the hologram also is a color filter for other colors, but ordinarily you see the image only in red. The reason is twofold. First, the intensity of the Bragg-reflected light is much less for colors other than red, and second, the range of angles over which the red light is reflected overlaps the Bragg angles for other colors. Other colors are submerged under the red reflected light.

⁴See Chapter 11 for a discussion of true-color.

A reflection hologram does not produce only a red image with white light illumination. Some can be moved about in the beam until eventually an image is seen quite distinctly in some other color. For example, there may be a main image of a bright red rose, and a secondary image of the same rose in a blue-green color. For this to happen, the Bragg angle for this other color must be so greatly different from the red light that at most only a small residual red reflection occurs. This leaves the other color visible.

A reflection hologram viewed with white light often gives a greenish image rather than a red one. This effect is quite different from the color effects just mentioned. It is due to shrinkage of the emulsion during the chemical processing. This shrinkage reduces the spacing between the Bragg planes, so that the hologram appears to have been made with light of a shorter wavelength than was actually used. The nature of the emulsion medium and the processing usually shift the effective wavelength to the green spectral region. The image in such cases has a greenish tint. Even with shrinkage, the hologram can still be viewed with the original red laser light, but the central Bragg angle will be different from how it would be if there had been no shrinkage.

MORE ABOUT IMAGES WITH REFLECTION HOLOGRAMS

Early in this chapter a virtual orthoscopic image and a real pseudoscopic image were discussed, and how they can be seen with a reflection hologram. Here we will discuss briefly the production of these images. Figure 6.7 shows a simplified conception of the hologram, with the partially reflecting surfaces curved and essentially parallel. Part (a) shows the usual configuration for exposing the emulsion, with the film relatively near the object, and the general shape of the interference fringes and how they lie in the emulsion. As (b) stresses, the finished hologram is different as seen from the side of face A from how it is seen from the side of face B. The plate "remembers" the orientation it had when it was made.

If the plate is reinstalled in its position in (a), with the illuminating beam coming from the left and with the original object removed, the light incident on the plate will encounter surfaces that are convex toward the source. The reflected rays may then diverge as they move back toward the left, and this would correspond to production of the virtual image. But this cannot be the whole story, for the reflection hologram does not simultaneously produce a real image, although a transmission hologram does (in which the readout beam may also fall on convex surfaces). The Bragg effect must be invoked for a fuller explanation.

Figure 6.5 shows how a stack of flat and parallel surfaces can produce a strong Bragg-reflected beam (for a given wavelength and surface spacing) at only one angle. If in looking at the figure you imagine the surfaces to become

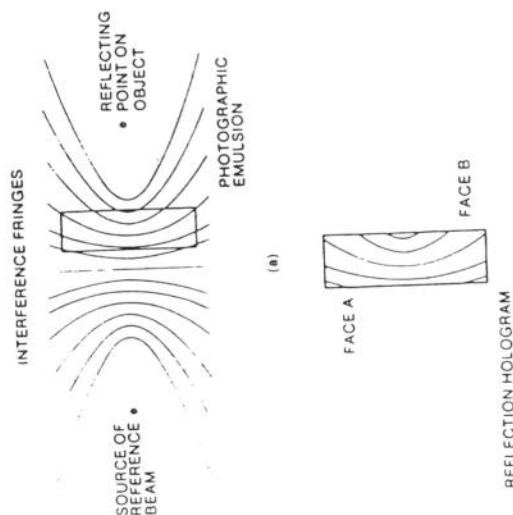


Figure 6.7. (a) Schematic representation of a setup for making a reflection hologram. (b) The finished plate contains surfaces that are different when viewed from the side of face A and from the side of face B.

curved gently, convex upward, you can understand why the reflected rays will diverge, and accept that the condition for strong Bragg reflection still cannot be met by any other rays (such as would be required for a set of convergent rays).

If instead you imagine the surfaces in Figure 6.5 to become curved gently, and concave upward, you can see how the reflected rays will be convergent and why only these real-image rays undergo strong Bragg reflection.

Now let's try a simple experiment that shows another interesting property of some reflection holograms. All you need is a reflection hologram that does not have some opaque backing on one side, so that you can shine light through it. The experiment consists of using the reflection hologram as if it were a transmission hologram. That is, you face the source of white light or the laser (with a spread-out beam), interpose the hologram between your eyes and the source, and look through the film. You should be able to see the virtual image. It will be fainter than the image you see when you use the plate normally as a reflection hologram, because strong Bragg reflection still takes place, and that back-reflected light is not available to your eye to see the virtual image. If the

hologram gives only a somewhat bright image in the reflection mode, the transmission mode image may be quite bright, but if your hologram produces a very brilliant reflection mode image, you may see only a very faint and indistinct transmission mode image.

The image will be seen on the other side of the plate, but probably not very far over there because reflection holograms are usually made with the object near the plate. If you use a white-light source to view the transmission mode image, you will see it in a color other than red. We will call the color a bluish ashen color. It is the result of the reflection of much of the red component from the white light from the source, together with some of the green (due to shrinkage of the emulsion). What colors are reflected back are subtracted from the light available for viewing the image. If your hologram gives you a viewable virtual image when used as a transmission hologram, you probably will also be able to see a real image. It should be pseudoscopic, fainter than and colored differently from the pseudoscopic reflection mode image.

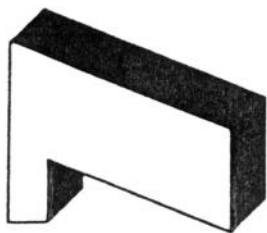
This use of a reflection hologram as a transmission hologram is an instance in which properties of both kinds are present in one plate. (More about this in Chapter 7.)

A SPECIAL FEATURE OF NEAR-SCENE REFLECTION HOLOGRAMS

A transmission hologram contains optical information about each point on an object in the scene spread out over the whole hologram. This accounts for the property of redundancy. But a reflection hologram made with the object close to the emulsion can form the interference fringes that pertain to any one point on the object only in a small region that is near to the object point. The optical information about the point is located in nearly one-to-one correspondence with the object points.

In simpler language, the result is that light shone onto the plate to view the image is reflected from the plate rather as it would be if it were shone onto the original object—point by point—whereas the whole plate contributes to the image of each object point in the case of a transmission hologram.

There is another kind of hologram that also has this property. It is called a *projection* or *stand-out* hologram, and it is discussed in Chapter 10.



Phase holograms and some other types

INTRODUCTION

Transmission and reflection holograms are two contrasting types. In the world of holography there are still other pairs of contrasting kinds: surface and volume, some are bleached and unbleached, thick and thin. Such distinctions and their significance are discussed in this chapter.

PHASE HOLOGRAMS AND BLEACHING

In ordinary photography light falling on the emulsion activates the silver halide grains, strongly where the light is intense and less so elsewhere. When the negative is made there are regions that are quite dark (those exposed to high light intensities) and less dark regions, shading down to virtually transparent ones. In holography the situation is similar, yet the entire hologram appears dark. The reason is that the interference fringes are very closely spaced and extend over the whole film.

Obviously such a hologram cannot transmit light very well. It is appropriately called an *absorption hologram*. The *efficiency* of a hologram is defined as the fraction of the incident light that is incident on it that is actually used in reconstructing the images. For a transmission hologram with a typical thickness

of several microns, the efficiency may be only 2 or 3 percent! For reflection holograms the efficiency may be only 5 to 7 percent.

The situation is much better with *phase holograms*. Phase holograms of both the transmission and reflection types can have efficiencies that approach 100 percent. With so much of the darkness that absorbs light absent, a phase hologram can put more of the available light into the image, which will be more brilliant. The improvement is so great that it is little wonder that most finished holograms today are phase holograms.

Some phase holograms are made with holographic media of special kinds that do not blacken. An example is a kind called *dichromated gelatin*. When this substance is exposed to light and processed by special means, the optical information in the form of interference fringes is recorded, but by means of variations in the "index of refraction" throughout the medium, which remains transparent. The following section will discuss this effect further. Another example of a special phase hologram medium is *thermoplastic*. This reacts to the heat energy carried by light incident on it. Still other phase media are known.

Phase holograms are usually created by converting an ordinary absorption type of hologram made with ordinary silver halide emulsion into a phase hologram by a step in the processing of the film. After the film has been developed, but before the processing has been completed, the film is soaked in a chemical solution to bleach it. The **bleaching converts the dark grains in the negative into other silver salts that are transparent**. The bleaching does not erase the interference fringes but rather transforms them.

BLEACHED HOLOGRAMS AND THE GEOMETRIC MODEL

Using silvered mirrors is part of daily life, so you probably understand the reflection of light from opaque reflecting surfaces. Because of this, the idea of reflection from the dark hyperbolic surfaces in a hologram is easily accepted. But how can the reflections be understood in the case of a bleached hologram?

Reflection from clear surfaces actually is also common in daily experience. For example, you can of course see yourself reflected in a store window, often with very high efficiency. Again, you can see that there are bubbles or other defects in a sheet of glass, although they are not opaque.

Technically such reflections are said to take place because the light encounters regions where the "index of refraction" changes. In Figure 7.1 there is an interface between two media, which might be air and glass, or a clear oil on water, for examples. Let's assume that the media are both solid, since this corresponds to transparent fringes in the body of an emulsion. The speed of light generally varies according to different media, and is less than in vacuum or in air. For example, the speed in one kind of glass may be $2/3$ as great as in vacuum. For another, the speed in a plastic may be $3/5$ as great as in vacuum. In any case, the *index of refraction* for a medium is defined as the ratio of the



Figure 7.1. Light falling on the interface between two optically different media generates Huygens' wavelets, which spread out at different speeds upward and downward.

speed of light in vacuum to the speed in the medium. In the two examples given, the index is 1.50 in the first case and 1.67 in the second.

It follows that if light travels initially in one medium and then encounters an interface with another medium that has a different index of refraction, the speed of the light will change when it crosses the interface. In Figure 7.1 a ray of light does just that. According to Huygens, the point on the interface where the ray arrives becomes a source of spherical wavelets. Some of these spread upward from the point, and some spread downward. Because of greater light speed in the upper medium in the case illustrated, each wavelet will travel farther in a given time interval than will the corresponding wavelet in the lower medium.

In Figure 7.2 this is applied to the case in which initially there is an extended beam, rather than a single ray. The various wavelets traveling up from the interface have an envelope, and this is the *reflected* wavefront. The wavelets traveling in the lower medium also have an envelope, and this is the *refracted* wavefront. More detailed analysis shows that the reflected beam forms an angle at the interface that is equal to the angle formed by the incident beam, just as is true in reflection from a silvered surface such as an ordinary mirror. The refracted beam changes its direction from that of the incident beam.

In this simple way you can perceive the physical basis for saying that light undergoes reflection and refraction when there is a change in the index of refraction. In a bleached hologram, or in any phase hologram, it is the great reduction in loss of light through absorption that is important, and the optical information in the interference fringes remains. This information is of two kinds: phase and amplitude information. (See Chapter 4.) The amplitude information is recorded by variation in the index in the medium, for these variations affect the reflectivity. This is in lieu of variations in reflectivity because of varying degrees of darkness in the partially reflecting surfaces in absorption holograms. Phase information is recorded in the *shapes of the regions*, as in absorption holograms, because the shapes depend on phase relations between the reference beam and the object beam when the hologram is made.

surface takes on a texture that can be described roughly as rippled. The details of the ripples constitute a surface hologram.

If the surface is coated very thinly with reflecting material to retain the surface texture, the image can still be viewed in reflected light. It may even be brighter than before, but because a surface cannot contain the wealth of optical information that a volume can, such an image can be expected to be of inferior quality.

Notice that the bleaching of a holographic emulsion can also change the emulsion thickness, and produce the effect just described. An interesting consequence of this is that it makes possible the mass production of copies of holograms. After the hologram is duplicated, the surface is plated with metal. Then the metalized surface is used to make imprints into plastic, much as a master phonograph disc is used to stamp out numerous copies.

THICK AND THIN HOLOGRAM

A thick hologram can be defined by considering a reflection hologram. The hologram is thick if the emulsion depth is great enough to contain a substantial number of Bragg-reflecting surfaces. If it can contain only a very few surfaces, the hologram is thin. This is not a very precise definition, but an emulsion roughly ten microns deep, which would contain some tens of surfaces (assuming red laser light), is considered thick. An emulsion only about one micron deep, which would contain only one or two surfaces, is considered thin. Because a strong Bragg effect is desirable for a reflection hologram, a reflection hologram should be thick.

For transmission holograms the distinction between thick and thin has to do with the sizes of the reflecting surfaces that are present between the faces of the hologram. This is illustrated in Figure 7.3. In (a) a thick transmission hologram is shown. In (b) the contrasting case of a thin one is shown. In the ultimate case of an extremely thin emulsion, only striations that lie on a surface can be present. This would be considered a surface hologram.

Many of the properties of thick transmission and reflection holograms can be understood using the geometric model plus Bragg reflection. Yet the geometric model is not capable of explaining thin holograms. As Dr. Jeong says:

As is true with all models, our geometric model for holography must break down at a limit. This limit is the "thin" hologram. A hologram is considered thin when the separation between the hyperboloidal surfaces exceeds the thickness of the emulsion. At this point our model will start giving the wrong answers. For example, a thin reflection hologram according to the model should give a white image when illuminated with white light, while in reality no image is formed in this way. Also, since there is no longer any Bragg effect, a transmission hologram should also be given a white image if white light

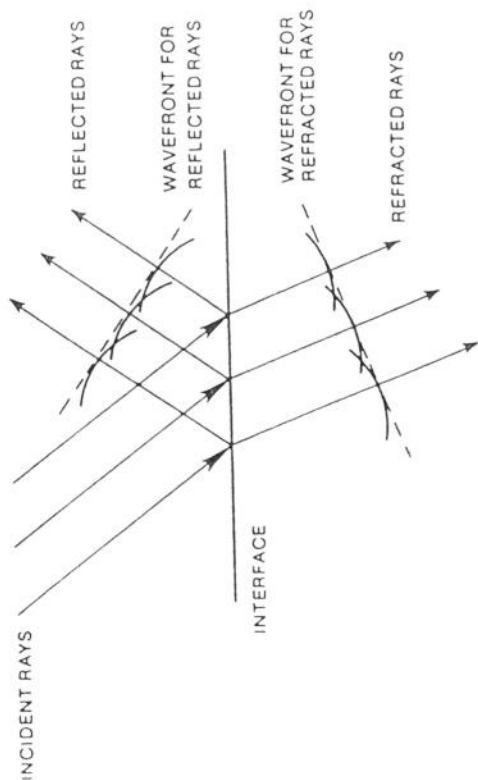


Figure 7.2. Some of the incident light is reflected and some is refracted. The angle of reflection is equal to the angle of incidence. The angle of refraction is different, and such that the refracted beam "bends" toward the normal to the interface.

SURFACE HOLOGRAMS

In the discussion above it was taken for granted that the interference fringes are recorded throughout the volume of the hologram. Actually, holographic information can also be recorded on the surface of a medium. An important case is a technique in acoustic holography in which the fringes appear as ripples on the surface of a body of water. (See Chapter 11.) In another method, an oil film is scanned by a modulated electron beam, producing deformations of the surface of the film. In still another method, thermoplastic is used.¹

You don't need to resort to such exotic techniques to get a surface hologram. In fact, when a hologram is being made in the usual way with an ordinary silver halide emulsion, during the fixing process residual undeveloped grains are removed from the emulsion, and there is partial collapse of the substance, greater and lesser in various places. As a consequence of this interior effect, the overall thickness of the emulsion changes from region to region, and the

¹Details may be found in *Optical Information Processing and Holography* by W. T. Cathey (New York: John Wiley & Sons, 1974, pages 141 ff). This is also a good source of information about the differences between surface and volume holograms in general.

THE TRANSITION FROM TRANSMISSION TO REFLECTION HOLOGRAMS

The distinction between a transmission and a reflection hologram is usually very sharp. In the first case the sources of the reference beam and the object beam lie on the same side of the plate, as in Figure 4.1. For a reflection hologram, the configuration is more as in Figure 4.12, with the plate essentially between the light source and the object. These or even more distinct configurations are often used and result in holograms easily classified as one kind or the other. Yet often an arrangement is used in which the locations of the source, object, and plate and the orientation of the plate make the matter of classification less clear. As already noted, a hologram may be one kind and yet show properties of another. (See Chapter 6.)

Figure 7.4 shows some sample configurations out of the infinite varieties possible. In (a) point source S and small reflective object O are used to make a transmission hologram. Because S and O are very close together, the hologram will be nearly *in-line*. This means that when the image O' is reconstructed as in part (b), you have to look almost back into the laser beam to see the image. In this case the hyperbolic surfaces are essentially perpendicular to the faces of the plate, and the Bragg effect is virtually absent.³

In (c) source S has been moved so that the reference beam comes to the plate more obliquely, and the surfaces are tilted in the plate. When this off-axis hologram is reconstructed, a ray in the reconstructing beam can encounter a number of surfaces along its path, as shown in (d). Because of this, the hologram could be called a Bragg-effect transmission hologram.

In (e) there is a clear configuration for a reflection hologram. The surfaces now lie nearly parallel to the faces of the plate. This is favorable for Bragg reflection during readout of the image. In (f) the conditions for Bragg reflection have been optimized, though at the cost of in-line viewing. To avoid this, a configuration between those in (e) and (f) would be used, but more like the latter, while Bragg reflection would still be predominant.

The preceding discussions considered cases in which the properties of one kind of hologram could be detected with a hologram nominally of the other kind. Let's conclude this section with an example of how these concepts can be useful. Consider the following statement: "The importance of intermodulation noise is not great with white light reflection holograms." Once the meaning of *intermodulation noise* is understood, this becomes readily decipherable. Figure 7.5 shows the making of a reflection hologram of an object. Source S and the object are on opposite sides of the plate, and such surfaces as the one labeled

³Sometimes such a hologram is said to be *plane* because the volume of the emulsion is not being used to the full. Other types would then be called *volume* holograms. Unfortunately, the technical terminology in holography has not been universally agreed on, so you must watch out for usages that differ from what you might expect.

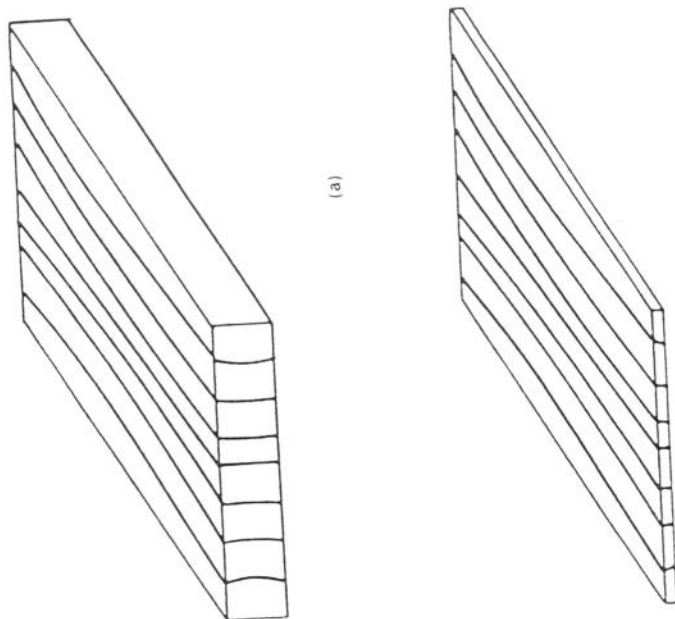


Figure 7.3. (a) A thick transmission hologram has substantial portions of the hyperbolic surfaces in its volume. (b) A thin transmission hologram essentially has only striations that lie on its surface.

is used for the reconstruction. The fact is that the image will be smeared in a continuous spectrum.⁴

You may be thinking that extremely thin holograms are of much less general interest than are thick holograms. This is true, so the breakdown of the geometric model in these special cases is not ordinarily a matter of concern. That is not the whole story, however, for some properties of thick holograms also elude the explanatory powers of the geometric model. These properties, and those of thin holograms, will be dealt with in Chapter 9.

⁴T. H. Jeong, "Geometric Model for Holography," *American Journal of Physics*, August 1975, page 717.

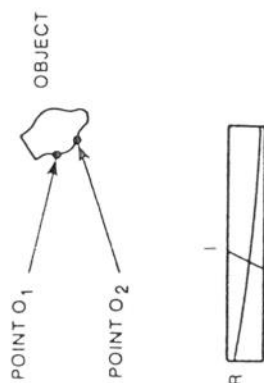


Figure 7.5. The interference surface R is the kind wanted in a reflection hologram. The surface I (formed by interference between beams from two points O_1 and O_2 on the object) is unwanted.

R lie along the length of the plate. But rays from any two points such as O_1 and O_2 on the object will interfere in the plate and produce interference fringes. These fringes are unwanted, and if their consequences are visible in the reconstructed image, they produce "intermodulation noise." However, the fringes produced by O_1 and O_2 will be oriented in the plate much as in the case of the one labeled I . Such fringes as I will constitute a transmission hologram, and with fringes such as R a reflection hologram. If the hologram is viewed as a reflection hologram, with a readout source at point S , the fringes of the kind represented by I will throw the light in directions far from that used to view the image, and the intermodulation noise will not be troublesome.

SOME MORE TERMINOLOGY

In the literature of holography you will hear of *Fresnel* and *Fraunhofer* holograms. A *Fresnel* hologram is produced when the object being holographed is near the plate, so that the object beam wavefronts are spherical. A *Fraunhofer* hologram is produced if the object-to-plate distance is so great that the object beam wavefronts are virtually plane. The details of the interference pattern recorded in the plate will be different in the two cases for a given reference beam. Our purpose here is only to say enough about the phraseology to keep

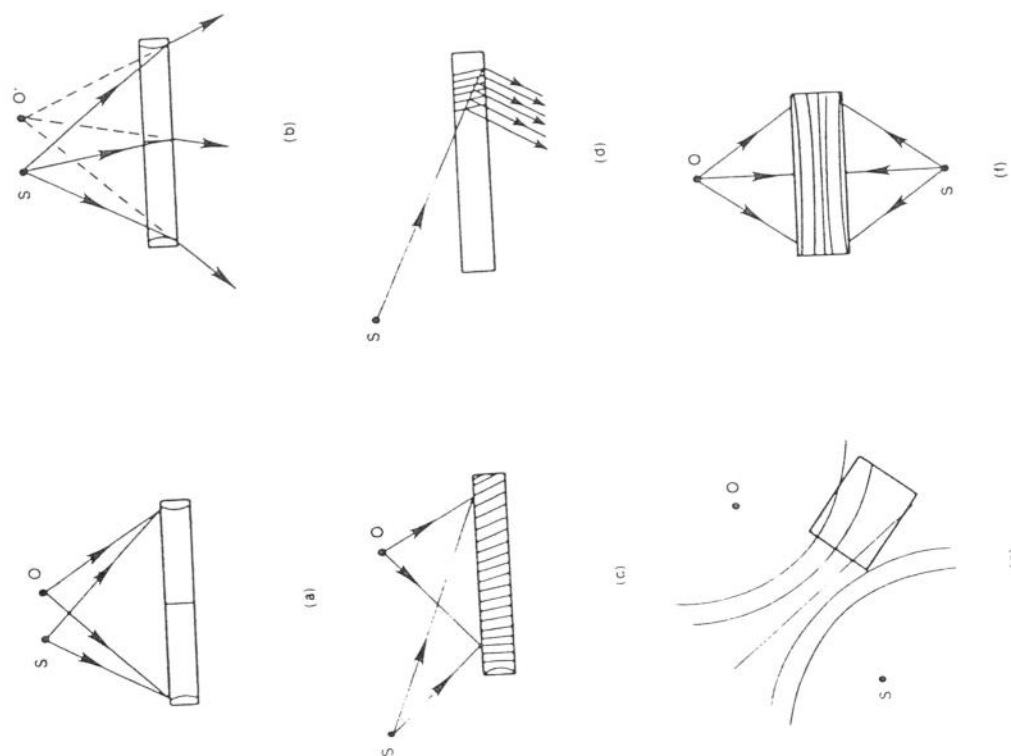


Figure 7.4. (a) Making a nearly in-line transmission hologram. (b) Reconstruction of the virtual image. (c) An off-axis configuration. (d) How Bragg reflection can occur to some degree with a transmission hologram. (e) Making a reflection hologram with an off-axis arrangement. (f) The in-line reflection hologram configuration.

you from being baffled when you encounter the terms. Further discussion will be found in Chapter 9.

Now let's briefly consider what is meant by a *Fourier* hologram. Consider a small light bulb. Light from the bulb that falls on a screen is unfocused light, and an image is not formed. If a converging lens is placed between the bulb and the screen and the distances are adjusted suitably, a focused image can be formed. Once this condition is met, if the screen is moved closer to or farther from the lens, there will be only a defocused blur. Yet optical information about the source is present in the light beam. It is said to be in the form of a *Fourier* transform. If the beam is now allowed to fall on an emulsion, and a coherent reference beam is also shone on the emulsion, an interference pattern will be formed. The developed plate will be a *Fourier* hologram. It cannot be viewed merely by shining the original reference beam on it, but a lens can be used to reconstruct the image. This kind of holography is an exception to the common characterization of holography as lensless photography. Fuller discussion and an application are given in Chapter 11.

The phrase *lensless Fourier* hologram refers to a different use of the word "Fourier." This is a transmission hologram made with the source of the reference beam and the object being holographed in the same plane. Such a hologram has only some properties of a true *Fourier* hologram, so the title is a misnomer.



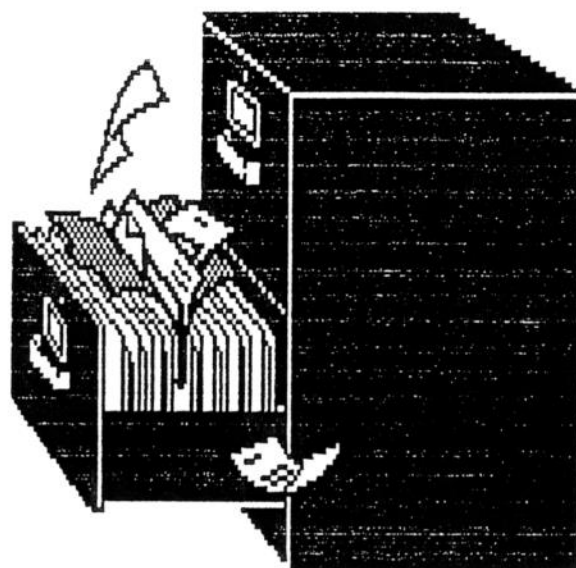
Some basic setups for making holograms

INTRODUCTION

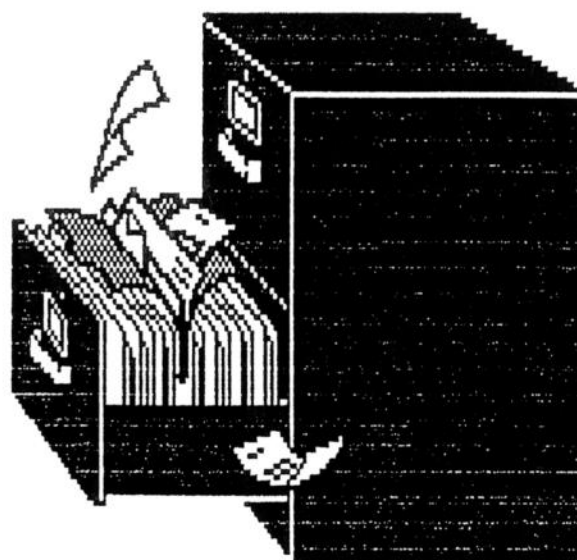
This chapter describes a number of ways to make transmission and reflection holograms. If you want actually to make holograms you will of course need practical how-to-do-it information as well. Chapter 13 discusses the material requirements in much greater detail than here. Chapter 14 takes up the making of holograms in the style of a laboratory manual.

THE ESSENTIAL APPARATUS AND FACILITIES

You don't need a lot to make holograms. The permanent items are a suitable laser; a table that is shakeproof to holographic standards; a few simple optical components (small lenses, front surface mirrors, and a beam splitter or two); some mounts to hold the optical components in place on the table; and a few very simple ancillary things, such as a block of wood to serve as a shutter, pieces of cardboard to screen off unwanted light, and trays to hold processing chemicals. The expendable items are photographic film or plates suitable for holography and chemicals for developing and bleaching the emulsions. Finally, the table has to be housed in an area that can be darkened when a hologram is being made, and you need an area that can be darkened during processing of the film.



SEPARATOR



the amplitude of the transmitted light in the reconstruction process and does not affect the optical phase of the light, and (2) the diffraction effects leading to the reconstructed image can be regarded as taking place in a very thin layer, essentially within a plane. These assumptions are valid for many, but not all, holograms.

The first distinction we can make between different classes of holograms is in the way in which light is diffracted by the hologram. Thus far, we have discussed only *absorption* holograms, wherein the *amplitude* of the incident light wave is modified in the reconstruction process through absorption by dark areas in the developed emulsion. A second type of hologram is the *phase* hologram, wherein it is the *optical phase* of the incident light that is modified in the reconstruction process, the amplitude of the light remaining unchanged. A phase hologram can be obtained by recording and developing a conventional absorption hologram and then bleaching it. In the bleaching process, the silver in the developed emulsion is replaced with some transparent salt of silver. This salt, silver bromide for example, has a refractive index different from the supporting gelatin matrix. As the incident light waves travel through the bleached emulsion, the local variations in refractive index introduce corrugations in the reference wave and, in the process, produce a replica of the original object wave. In the absence of Bragg selectivity, which we shall discuss shortly, a bleached hologram produces not only the desired image on reconstruction, but additional "ghost" images as well. Unless the recording geometry is properly chosen, these ghost images can overlap or interfere with the desired image. The principal advantage of the phase hologram over the absorption hologram is in diffraction efficiency, which is defined as the ratio of light power diffracted into the reconstructed holographic image to the power illuminating the hologram. The maximum diffraction efficiency for an absorption sine grating is 6.25 percent, and even this efficiency is not obtained in practice. For a thin sinusoidal phase grating, on the other hand, it is theoretically possible to diffract as much as 34 percent of the incident light into one of the diffracted beams. With the same illumination, a reconstructed phase hologram image is generally much brighter than its absorption hologram counterpart.

The second distinction we make between holograms relates to the thickness, or volume, of the holographic emulsion. Consider the diagram in Fig. 7.13. Plane waves interfere with spherical waves from source S in the region shown. The dark, curved lines show the locations of the interference maxima—the positions of greatest exposure for a photographic emulsion that intercepts the interfering waves. Holographic plates are shown in four different locations. Location ① is known as the *in-line* position and is the position of the holograms recorded by Gabor in 1948. Note that the fringes are widely spaced, suggesting the suitability of this location for recording holograms with low-resolution (and hence more sensitive) films. The disadvantage of the in-line location is that the reconstructed image is not separated from the undiffracted light and is consequently difficult to view.

Location ② is typical of hologram placement for *off-axis reference-wave* holograms, the kind we have discussed in earlier sections. For small angular offset (the average angle between the reference wave and the object waves) and for emulsions that are not too thick (several micrometers or less), the interference fringes exposing the hologram will have a separation large compared to their extent in the thickness of the emulsion. Holograms of this kind are referred to as *plane off-axis* holograms.

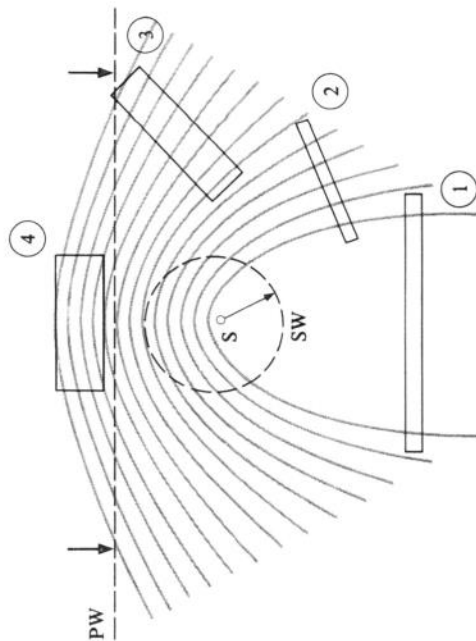


Fig. 7.13 Interfering plane wave (PW) and spherical wave (SW), represented by dashed lines, produce different grating characteristics in the holographic emulsion. Light lines show loci of interference maxima, where film exposure would be greatest. Thick and thin holographic emulsions are shown in four locations (see text): ① on-axis, ② off-axis, ③ Bragg-effect, and ④ white-light.

As the angle of the reference wave increases to around 90° , as in position ③, the spacing between the exposing interference fringes becomes quite small. The silver remaining in the emulsion upon development of the hologram forms closely spaced and partially reflecting planes that diffract light on reconstruction only if the playback reference wave is incident on the hologram in a narrow range of angles about the original angle of incidence. For an emulsion 12–15 micrometers thick, for example, such a collimated reference-wave *Bragg-effect* hologram*

* Named after W. L. Bragg, who first discussed such angular selectivity in connection with x-ray crystallography.

can be reconstructed only if the reference wave is incident at an angle within about 2° of the original recording angle.

When the holographic plate is exposed in position (4), the interference fringes are parallel or nearly parallel to the emulsion. Assuming the emulsion is sufficiently thick to record more than several dozen fringes in its volume, the processed emulsion will serve not only to reconstruct the hologram, but also as a selective interference filter, filtering out all but a narrow band of colors. Holograms recorded in this way are often referred to as *white-light holograms*,* because they can be reconstructed with white light. The hologram itself, in filtering the light, introduces the necessary degree of temporal coherence, or monochromaticity. (The light source used to reconstruct the hologram must still exhibit good spatial coherence, and should therefore be compact, approximating a point source. Flashlights and streetlamps are often adequate; also the sun is often an excellent source.)

Another type of white-light hologram is the *multiplex hologram*, which can reproduce motion as well as depth. The construction process begins by filming with a motion picture camera the action to be shown in the hologram. It is necessary that there be a relatively circular motion of the camera about the subject. This is achieved either by moving the camera on a circular track or by placing the subject on a rotating table. The motion picture film is developed and a multiple hologram is made by imaging single frames of the film in sequence along with the reference wave onto a holographic film. To compensate for the distortion caused when the finished hologram is bent into a cylindrical shape, a cylindrical lens is inserted in the object beam path, compressing the image onto a $20\text{ cm} \times 1\text{ cm}$ slit mask in front of the holographic film. The film is moved in small increments perpendicular to the slit length, and other overlapping holograms are made of the succeeding frames which give information about the subject (1) from slightly different angles and (2) at different times. The developed and bleached hologram is bent into a cylinder and illuminated by a small line source along the cylinder axis. Because the relative motion of the camera has been translated into different angles on the holographic film, there is an illusion of depth, as there is with a conventional hologram. But, in addition, as the viewer walks past the hologram to observe new angles, additional frames are seen, giving the illusion of motion. Because of the nature of the construction process in which the real image is formed at the holographic film, the hologram may be constructed with white light despite the fact that the object/reference wave angle is less than 90° . Someday, when newer techniques have been developed, motion pictures based on holographic principles may be possible. For the present, Hollywood can only use holograms as props.

* Also known as Lippman-Bragg, reflection, or Denisyuk holograms, soviet physicist Y. N. Denisyuk being the inventor in 1962.

Before leaving our discussion of hologram types, it is appropriate that we consider briefly one final class of holograms, those generated by computer, or *computer-generated holograms*. If an object can be specified mathematically, it is possible for a computer to calculate the locations of interference fringes as they would be recorded on a photographic plate if this object were placed in a particular hologram recording setup. These fringes, or their equivalents, can be plotted with an x-y plotter, and the resulting pattern photoreduced to a scale where it can be illuminated with the assumed reference wave. The object, which may never have actually existed, is "reconstructed" and can be examined from a number of perspectives. Computer-generated holograms can be recorded in several different ways. One large drawback is the great amount of computer processing time needed to generate the interference pattern. They have, nonetheless, an important role to play in holography, particularly in the area of optical data processing and in the testing of optical components.

7.3 APPLICATIONS OF HOLOGRAPHY

Holography has found application in a number of areas, and new uses are being developed all the time. The novelty and attractiveness of full-perspective, three-dimensional displays have inspired the limited use of holography in the advertising

SPECKLE INTERFEROMETRY

In addition to holographic interferometry, there are a number of related, nondestructive testing techniques, known collectively as *speckle interferometry*, that make use of the random laser speckle pattern discussed in Section 2.5. One such technique is *time average speckle interferometry*. The object to be tested, say a vibrating diaphragm, is made optically rough (with flat white paint, for example, to produce a good speckle pattern), illuminated with laser light, and photographed using high-contrast photographic film. At the film, the speckled image distribution is allowed to interfere with a reference wave that has been introduced by means of a beamsplitter. As the object vibrates, those regions of the image corresponding to the vibration nodes stand out as patterns of stationary, high-contrast speckle. In other regions, however, the speckles move about and become blurred due to the movement of the diaphragm. The bright, stationary speckles satisfactorily expose the film. The average irradiance of the smoothed-out portion of the speckle image is too low, however, to produce an adequate exposure. The result, on development, is a photograph of the object that shows clearly and with microscopic precision the regions of the diaphragm that were stationary and those that were moving.

Speckle interferometry, with its many variations, can be used with great versatility to measure object vibration, translation, strain, and rotation. Considering that speckle is often considered a nuisance in holography, it has clearly demonstrated redeeming features in other fields. The interested reader is referred to Ref. 7.10.

business. Of greater technical interest has been the use of holography in the high-accuracy measurement of object dimensions, deblurring of photographic images, and the characterization of low-density particulate matter. One especially interesting application has been in microscopy, where a hologram is made of an entire large-volume sample, freezing in one instant a record of many organisms suitable for later inspection at various depths under a microscope.

It is not possible to go into all the applications of holography, because the field is quite extensive and constantly growing. We shall look, however, at three important areas of application: (1) holographic interferometry, (2) holographic optical elements, and (3) holographic optical memories. Each of these applications exhibits a different aspect of holographic principles.

Holographic Interferometry

Holographic interferometry is a technique for measuring small displacements. (For an alternative technique, see box.) In one form or another, it can be used to observe and analyze, either in real time or after some delay, the microscopic flexure of a loaded support beam, to observe the strain in a fractured bolt, to record the shock wave from a bullet, to observe the vibrational modes of a kettle-drum head, or to detect hidden flaws in an aircraft tire.

The basic principles can be illustrated with the example of *single-exposure*, or *real-time*, *holographic interferometry* as applied to stress analysis. In this process, a conventional hologram is first made of the object to be tested. The hologram is developed in the normal manner and then returned to the exact position it held during exposure.* The object and the hologram are now illuminated just as they were during exposure, so that the reconstructed holographic image falls exactly on the object itself. If the object is now stressed slightly by heating or loading with weights, for example, fringes appear on it. These fringes result from the interference between the light waves from the now-distorted object and the holographically reconstructed waves from the "former" object. Where the distortion is small, the fringes are few in number; where it is great, many fringes appear. A direct qualitative indication of object stress is thus immediately available. An example of such a system of fringes appears in Fig. 7.14, where the object, a metal bar that is solidly attached at the bottom to a massive base, is stressed at the top. In the figure, the reilluminated bar and its holographic image are viewed simultaneously. Note that the rate of change of the beam displacement is greater near the top, the fringes being more closely packed there. Although in general the problem is quite difficult, with special geometries it is possible for one to interpret the fringe pattern to obtain a direct, quantitative measurement of object deformation (and, indirectly, of object stress). See Prob. 7.12, for example.

* An alternative is to develop the hologram in place. This can be done if the holographic plate is held in a special glass-sided container, which can be filled with the processing solutions.

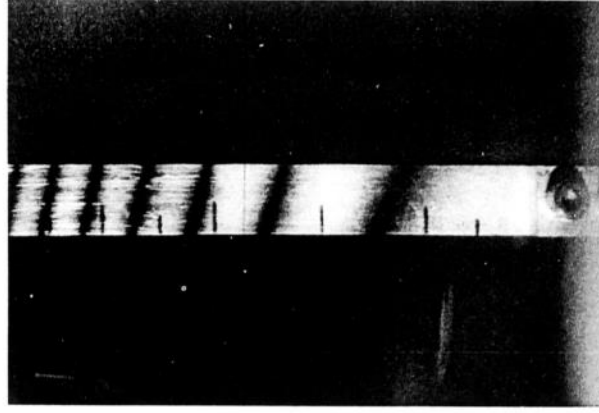


Fig. 7.14 Illustration of single-exposure holographic interferometry. Interference of the actual object wave and reconstructed object wave shows the amount of bending of the metal bar. The fact that the fringes are not horizontal indicates that besides the bend, the bar also suffered a slight twist about the vertical. The distance between marks on the bar is 1.27 cm (0.5").

A variation of real-time holographic interferometry, known as *double-exposure holographic interferometry*, has found widespread application in several areas of nondestructive testing. Nondestructive testing of tires was, in fact, one of the first large-scale industrial applications proposed for holography. The basic approach is similar to the real-time technique just discussed. In the present case, however, the original and the distorted objects are both recorded holographically in a double-exposure hologram. When the hologram is reilluminated with the reference wave, both images are viewed simultaneously, and the interference pattern representing the differences in the two recorded light distributions appears localized on the object. If the object has changed shape during the two exposures, even microscopically, the change will manifest itself as a system of closely-spaced fringes in the reconstructed image.

A common application is illustrated in Fig. 7.15, which shows the double-exposure hologram of a portion of the interior of an aircraft tire. In preparation for the hologram recording, the tire is placed in a chamber. After the first exposure, the chamber is pumped down to a partial vacuum. The second exposure is then

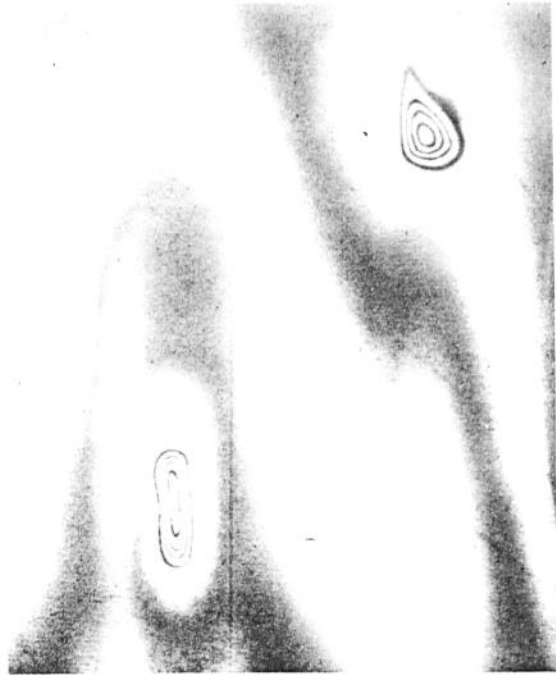


Fig. 7.15 Double-exposure holographic interferogram of an inside portion of an aircraft tire. Circular interference patterns indicate locations of tire ply separations. (Courtesy of Newport Research Corporation.)

made with the vacuum applied. If there is a local separation between the tread and the outer plies of the tire or between the various plies, both potentially dangerous flaws, air entrapped in the separation will cause a minute local expansion, a microscopic bulging, in the vicinity of the flaw. This bulging is readily discerned in the reconstructed hologram as a series of closely-spaced fringes. Commercial holographic testing systems developed for this purpose are able to test one tire every several minutes.

Closely related to single- and double-exposure holography is a technique known as *time-average holographic interferometry*. This technique allows the spatial characteristics of low-amplitude vibrations of an object—for example a drum head, speaker cone, or metal diaphragm—to be mapped out with great accuracy. Figure 7.16 shows a vibrational mode of a guitar. The fringes represent contour lines of equal-amplitude vibrations of the guitar surface; the brightest contours occur at the nodes, or stationary points, of the vibrating surface. This is to be expected, because light reflected from these points always interferes with the reference wave at the hologram in the same way, yielding a recorded fringe pattern of high contrast and, hence, high reconstruction amplitude. For other

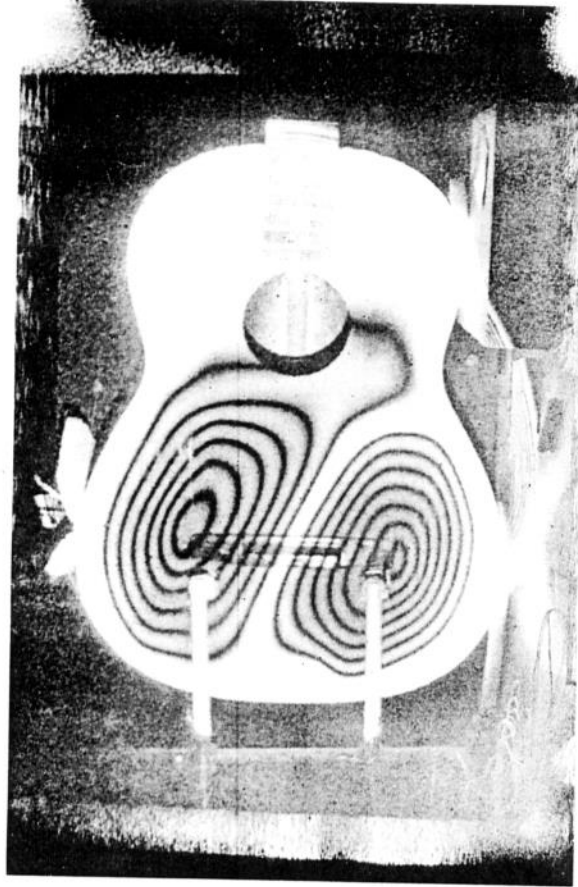


Fig. 7.16 Photograph of a time-average holographic image of a guitar. The fringe pattern shows the vibrational mode of the guitar at a particular frequency. (Courtesy of K. A. Stetson.)

regions of the vibrating surface, however, the corresponding interference pattern is moving and, for long exposures, produces a diffraction pattern of reduced or even zero reconstruction efficiency. The relationship between the pattern obtained and the vibrational amplitude at a particular location is not a simple one and is not derived here. It is possible, however, to use such time-average interferograms quantitatively.

Holographic Optical Elements

A hologram takes an incident light wave and, by means of diffraction, generates a different wave. In this sense the hologram can be viewed as a general kind of optical element with the capability of mapping one light distribution into another. The similarity between holograms and conventional optical elements has already been noted: A sine grating, the simplest of holographic records, changes the direction of a beam of light much as a prism does, and the sinusoidal zone plate behaves simultaneously like a positive lens and a negative lens.

This similarity between holographic elements and conventional optical elements can be exploited in a number of applications. Because of the strong wavelength dependency of the diffraction process (e.g., Eq. 7.6), holographic optical

elements are highly dispersive; that is, they behave differently at different wavelengths and must be used with monochromatic light or several of these elements must be combined in such a way as to eliminate wavelength-dependent effects. Nonetheless, holographic optical elements have a number of attractive features: They can be thin, lightweight, and compact; they can be easily fabricated and replicated; they can be very large in size. In addition, photographic emulsions can be coated onto arbitrary transparent surfaces for the fabrication of special-application, conformal optical elements. For example, one might manufacture holographic lenses that could be rolled up in tubes for storage or shipping.

One of the earliest and still most successful uses of a holographic optical element has been in optical spectrometers, where a high-resolution, holographically recorded grating replaces the considerably more expensive conventional grating. As holographic recording processes have been improved, so has the manufacture of these holographic gratings, to the point where holographic spectrometers now claim a substantial fraction of the spectrometer market. Similarly, holographic lenses (off-axis sinusoidal zone plates) have been used in imaging systems.

Just as glasses of different refractive indexes have been combined in the manufacture of lenses that perform equally well over a wide range of wavelengths, so it is possible to combine multiple holographic zone plates and gratings in a system suitable for achromatic imaging over most of the visible spectrum. Although the individual component elements must be very precisely positioned for them to function properly, achromatic holographic imaging systems constructed in this manner exhibit the attractive features of small mass and reproducibility. To replicate such a system, it is only necessary to duplicate several holograms, whereas with conventional systems, glass elements must be ground and polished and tested.

Holographic optical elements are often used in optical data processing systems. A particularly simple example is a coherent optical system (i.e., employing laser light) for recognizing alphanumeric characters. Let us consider in detail a subsystem for recognizing a specific letter—say the letter K. First, a hologram is made of a prototype letter K with type style and orientation appropriate to the recognition problem, as illustrated in Fig. 7.17(a). The reference wave is chosen to be a converging spherical wave. If the developed hologram is illuminated with the same spherical wave, the letter K is reconstructed. However, the converse is true also: If the hologram is reilluminated with light from the letter K, a spherical wave is produced that converges to a bright spot in the output plane of the optical system (Fig. 7.17b). There its presence can be detected with a photodetector. The light waves from other alphanumeric characters will also be diffracted by the hologram, but not into the suitably converging wave that is produced by waves from the correct letter (Fig. 7.17c). By detecting the presence or absence of a focused spot of light in the output plane of the system, one can determine whether a particular character is present in the input plane. Additional holographic optical elements can be arranged to allow an entire page of printed material to serve as

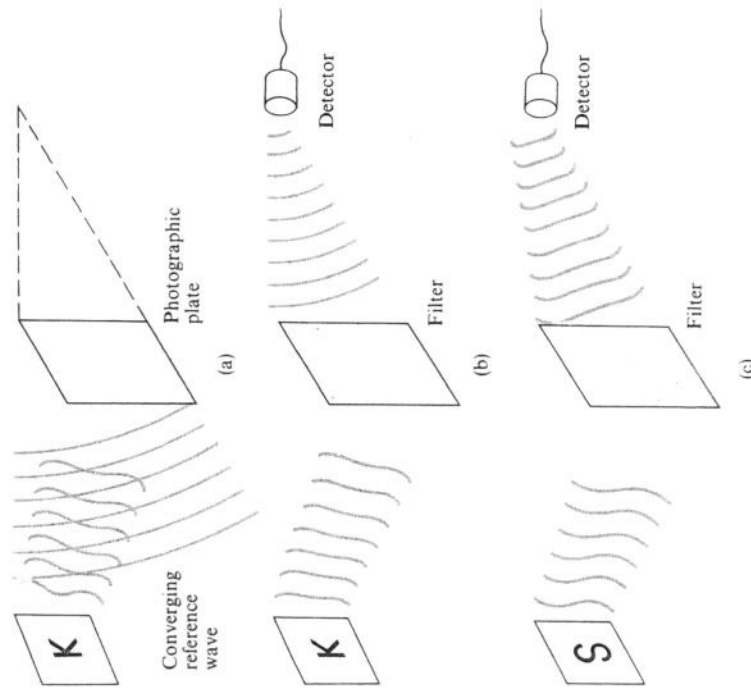


Fig. 7.17 Holographic character recognition (Vander Lugt filtering). The holographic filter, recorded as in (a), maps waves from a "matched" character into converging spherical waves, which are detected as points of light in the output plane (b). Waves from an incorrect character are also diffracted by the filter, but not into suitably converging waves (c).

input to the system. The locations of focused spots of light in the output plane then signify the locations of the corresponding "matched" character in the input plane.

Holographic Optical Memories

The use of photosensitive materials in large-scale memories to replace magnetic-tape or disk bulk storage systems on computers has intrigued scientists and

* The system is the optical analog of the so-called *matched filter* receiver sometimes used in radar signal detection. The holographic optical element is often called a *Vander Lugt filter* after its inventor, A. B. Vander Lugt.

engineers for some time. Optical memories are limited ultimately in their information storage capacity by the wavelength of light used to "read" or "write" the information. Assuming an argon laser operating at 488 nm is the light source, optical memories are theoretically capable of storing in excess of 10^8 bits of information per square centimeter of photosensitive surface, or nearly 10^{13} bits in a 1-cm³ volume. These numbers exceed the storage density limitations of magnetic systems by several orders of magnitude, and have made optical storage of digital information an area of extensive research. Much of the research conducted has centered on the difficult problem of preparing a photosensitive material with high resolution that can be written onto, read from, and erased with the fidelity of the recording process preserved over long periods of time.

The introduction of holographic techniques to the optical data storage area has led to optical memory systems for digital data that have several very attractive features. First, as noted earlier, holograms record information redundantly. Thus one bit of information, which in a conventional photographic recording process might be recorded as a transparent spot in a dark background, is in the holographic case recorded as a sinusoidal grating. The information for that one bit is spread out over the entire hologram. With conventional photographic recordings, a speck of dirt on the film could entirely block out the transparent spot corresponding to the bit. In the holographic case, a single speck of dirt has no significant effect on the recording. To recover the information recorded holographically, i.e., to "read out" the stored bit pattern, the holographic recording is placed in front of a positive lens and illuminated by collimated laser light. Each sinusoid generates a diffracted light beam, which is focused by the lens to a unique spot, there to be detected by a photodiode. The second advantage of the holographic recording of the bit pattern now becomes apparent, since the exact position of the hologram in the playback setup is seen to be unimportant. The hologram can be moved, either horizontally or vertically in the input plane, and the focused spots remain stationary. It is only necessary that the hologram be reasonably well illuminated. Binary information stored in sine-grating format thus exhibits simultaneously an immunity to "burst" errors (the speck of dirt) and an insensitivity to improper positioning in the readout system. These two characteristics have made holographic data storage attractive for moderately high-density storage of digital information when records are accessed frequently. Microfilm records of reports and journal articles are presently being made, for example, with cataloging information recorded in holographic form as well as in conventional form. Easy automatic readout of these data in a position-insensitive system is thus assured.

PROBLEMS

7.1 Referring to Eq. (7.1) describing the irradiance distribution resulting from the interference of two plane waves, under what circumstances does $I_o = I_m$?

7.2 Why must T_b and M in Eq. (7.4) satisfy the fundamental constraint $|T_b + M| \leq 1$?

7.3 The *diffraction efficiency* for a sine grating, η , is defined as the ratio of the light power in one of the diffracted beams to the power in the incident beam. In terms of the grating modulation constant, M , η for an absorption-type grating is given by $\eta = (M/2)^2$. Show that the maximum diffraction efficiency for a sine grating is 6.25 percent.

7.4 Following the approach leading to Eq. (2.24), show that Eq. (7.1) for the interference of two unequal-amplitude plane waves is correct. Assume that the two waves have amplitudes E_1 and E_2 . (a) What is I_o in terms of E_1 and E_2 ? (b) What is I_m in terms of E_1 and E_2 ?

7.5 For the case of normally incident plane-wave illumination, the angle of diffraction at a sine grating is given by $\sin \theta_d = \lambda f_g$. If the grating frequency, f_g , exceeds $1/\lambda$, then $\lambda f_g > 1$ and $\sin \theta_d > 1$, an impossible condition. Under these circumstances, no diffraction takes place. What is the maximum possible grating frequency, measured in grating cycles (or line pairs) per millimeter, if diffraction is to take place at a grating illuminated with the 514.5-nm light from an argon ion laser?

7.6 Referring to Fig. 7.4 for the T - ϵ curve representing a typical holographic emulsion, let α be the slope of the curve in the linear region, i.e., $\alpha = dT/d\epsilon$ in the vicinity of $T = T_b$. Assume that a grating is produced by exposing the film to the irradiance distribution of Eq. (7.1), and that I_o and I_m are chosen to restrict $\epsilon(x)$ to the linear portion of the T - ϵ curve. What is M , the resultant grating modulation constant, in terms of α ?

7.7 Figure 7.4 shows the T - ϵ curve for a typical negative-working photographic emulsion (where increased exposure yields reduced transmittance). What would the corresponding T - ϵ curve look like for a positive-working emulsion (where increased exposure yields increased transmittance)? Assuming that a suitable linear portion exists for a positive-working emulsion T - ϵ curve, do you see any reason why such an emulsion could not be used in the making of sine gratings and, hence, holograms generally? Justify your answer.

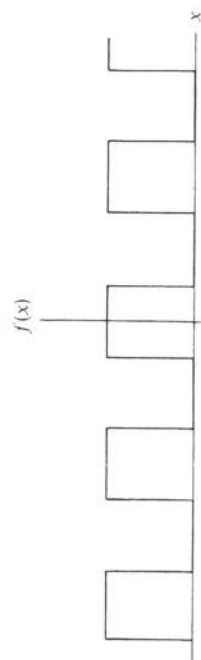
7.8 A square wave, such as the one shown in the figure below, can be represented by a Fourier series of the form

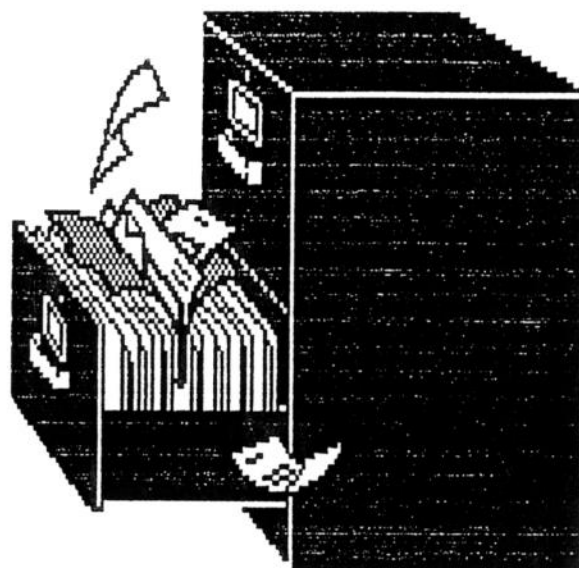
$$f(x) = b_o + \sum_{n=1}^{\infty} b_n \cos 2\pi n f_o x.$$

Show that a high-contrast diffraction grating that exhibits a square-wave transmittance profile, when illuminated with normally incident collimated light of wavelength λ , diffracts the light at multiple angles given by

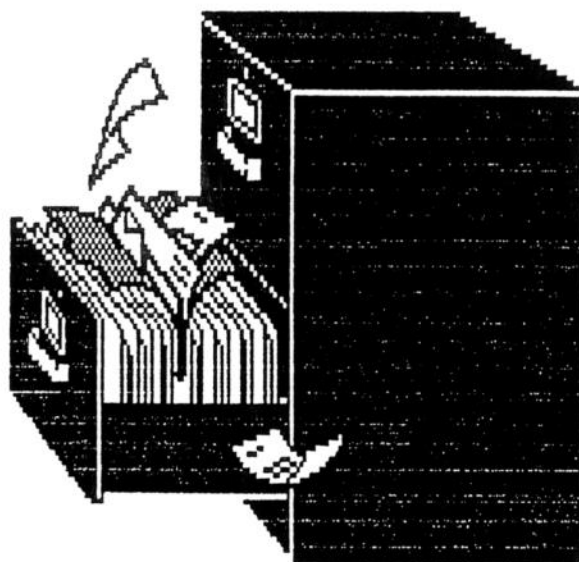
$$\sin \theta_n = n \lambda f_g$$

where θ_n is the diffracted angle associated with the integer n , and f_g is now interpreted as the fundamental grating period. What is the maximum allowable value for the integer n for which diffraction occurs, given λ and f_g ?





SEPARATOR



Simplified Processing Techniques for Holography*

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Silver halide emulsions for holography are processed in most cases by methods developed originally for general photographic work. If these are adapted to the specific needs of holography, much simpler techniques are possible. One such is a monobath for rapid *in-situ* processing of holographic emulsions in a liquid gate. This cuts processing time to around 120 sec and permits utilization of a very simple solution handling system for real-time hologram interferometry. Another is a reversal processing technique for volume phase holograms involving only two baths. This gives very low noise and high stability against exposure to light.

Photographic Science and Engineering 24: 105-107 (1980)

Introduction

Silver halide photographic emulsions are still the most commonly used material for recording holograms because of their relatively high sensitivity and ready availability from commercial sources. These ultrafine-grain photographic materials are processed in most cases by methods developed originally for general photographic work. If these are adapted to the specific needs of holography, much simpler procedures are possible. Two of these are described in this paper.

Monobath Processing

Typically, a problem encountered with conventional processing methods in real-time holographic interferometry is the appreciable delay between the time when the exposure is made and the time when measurements can start. During this period, the object can undergo dimensional changes. Processing can be simplified and speeded up by the use of a monobath in which development and fixing take place simultaneously.¹

Monobaths are usually compounded by adding a suitable quantity of a silver halide complexing agent such as sodium thiosulfate to an active developer formula. The visible image is formed by chemical development of the exposed silver halide grains. At the same time, the unexposed silver halide dissolved by the complexing agent migrates toward adjacent developing grains and is reduced there to metallic silver.^{2,3} Thus, a gross form of solution physical development also takes place. As a result, the silver deposited has a much more compact structure and, lower covering power. There is also an enhancement of contrast at edges, which can result in improved MTF even at relatively high spatial frequencies.⁴

Monobaths have not found much use in the past except for military applications, because of several problems such as reduced speed, high fog, and poor contrast. Trials with holographic emulsions showed that because of the high surface-to-volume ratio of the grains, very satisfactory results could be obtained with much lower concentrations of thiosulfate compared with those in monobath formulations for negative

materials. Low fog levels and a neutral black image could then be obtained even without an antifoggant.⁵

A typical monobath formula for use with Scientia 10E75 plates is shown in Table I, while the results of sensitometric tests with this formula are summarized in Fig. 1. This shows the curves of amplitude transmittance versus exposure for processing times ranging from 30° to 240 sec at 20°C. Very nearly the maximum speed and contrast are obtained with a processing time of 120 sec.

Even though the value of gamma obtained with monobath processing is distinctly lower than that for development in D-19 for 300 sec (3.6 against 5.4), the loss in contrast is mainly at higher densities, and there is little difference in the slopes of the T_a vs. E curves over the region used for holography. On the other hand there is an increase in speed of 50% with the monobath. This increase in speed was confirmed in practical trials. No significant difference in the brightness or contrast of the reconstructed image was observed.

One of the drawbacks of monobaths, especially in bulk processing of negative materials, is their tendency to deposit a sludge of metallic silver on standing. However, sludge formation with this relatively dilute monobath is negligible during the actual processing time. Sludging can therefore be avoided by using the minimum quantity of fresh monobath required for processing each plate and discarding it immediately after use.

This makes the use of a monobath very attractive for rapid *in situ* processing. A very simple gravity system, shown schematically in Fig. 2, is used to run measured volumes of either distilled water or the monobath from two reservoirs into the liquid gate, and the plate is exposed and viewed in water.⁶ Problems of index matching and differences in the emulsion thickness between exposure and viewing are eliminated, and real-time interferograms virtually free of residual fringes are obtained. These interference patterns are very stable and can be studied over long periods of time.

Reversal Bleaching

Conventional processing of photographic materials yields amplitude holograms which have relatively low diffraction efficiency. Much brighter images can be obtained with phase holograms.⁷ For this, as shown in Fig. 3(a), the silver image formed after developing and fixing is converted into a transparent salt, usually one of the silver halides, by one of a number of chemical bleaches.^{8,9} While the resulting phase holograms have improved diffraction efficiency, this has been

TABLE I. Monobath Formula

Stock solution A	
metol	3.5 g
sodium sulfite (anhydrous)	50 g
hydroquinone	15 g
potassium alum	5 g
sodium thiosulfate	50 g
distilled water to make	800 ml
Stock solution B	
sodium hydroxide	10 g
distilled water to make	200 ml

Mix 4 parts of A with 1 part of B before use. Use at $20 \pm 0.5^\circ\text{C}$.

* Paper presented at the Seminar on Holographic Materials and Processing, SPSE 32nd Annual Conference Boston, Mass., May 14-17, 1979. Received May 16, 1979.

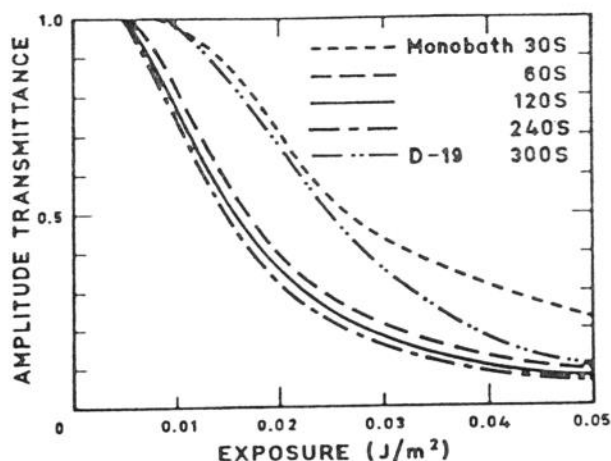


Figure 1. Curves of amplitude transmittance vs exposure for Scientia 10E75 plates processed in the monobath.

accompanied in most cases by a large increase in scattering and a consequent reduction in the signal-to-noise ratio. Very elaborate techniques to reduce scattering have been described in some publications.¹⁰

One major source of scattered light is nonlinear effects arising from the formation of a relief image at low spatial frequencies due to local tanning by the oxidation products of the developer.¹¹ If care is taken to minimize such effects, a significant cause of noise is the fact that the bleached hologram actually consists of discrete grains having a higher refractive index (n_2) embedded in a gelatin matrix of lower refractive index (n_1). The effective refractive index (n_{eff}) at any point is then, to a first approximation,¹²

$$n_{eff} = n_1 + 2\pi N n_1 [(m^2 - 1)/(m^2 + 2)] a^3 - i(4\pi/3) N (2\pi/\lambda)^3 n_1^4 [(m^2 - 1)/(m^2 + 2)] a^6 \quad (1)$$

where N is the number of grains per unit volume, a is their radius and $m = (n_2/n_1)$.

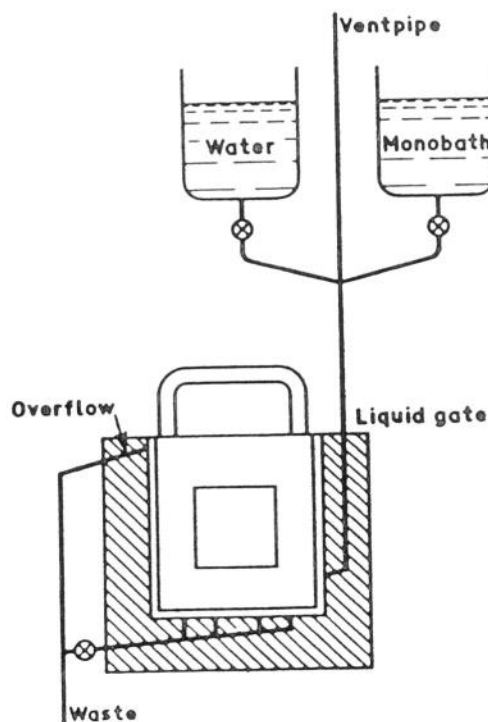
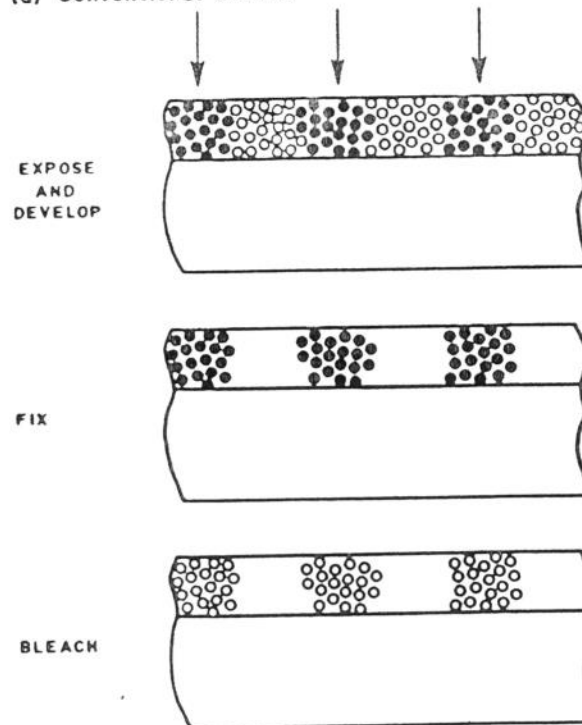


Figure 2. Solution handling system for *in situ* processing using the monobath.

(a) Conventional Bleach



(b) Reversal Bleach

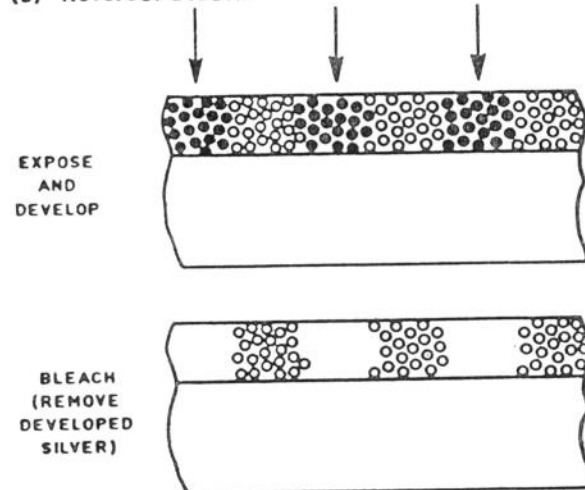


Figure 3. Steps involved in (a) conventional bleach techniques; (b) reversal bleach techniques.

Even if the grains are perfectly transparent, the refractive index contains, in addition to a real term which gives the phase retardation of the incident beam, an imaginary term which is due to the radiation reaction on the scatterers and corresponds to the extinction due to scattering. The fact that the loss is proportional to the sixth power of the radius of the grains highlights the importance of keeping grain size to a minimum in a bleached hologram.

On the other hand, during development the relatively compact silver halide crystals in the emulsion are converted into much larger filamentary aggregates of metallic silver. When these are oxidized to a silver halide by a bleach, there is a considerable increase in scattering.

A method of avoiding this is the use of a reversal bleach.¹³⁻¹⁶ Processing is simplified with such a bleach, since fixing is eliminated. As shown in Fig. 3(b), the developed silver is dissolved away, leaving a phase hologram made up of the undeveloped silver halide crystals. In addition, a further reduction

TABLE II. Processing schedule for reversal bleach

1. Expose in hologram set-up (recommended exposure for Scientia 8E75 plates, 1.0 J/m²).
2. Develop for 5 min (for developer formula, see below).
3. Rinse in distilled water for 3 min (2 changes).
4. Bleach for 5 min (for bleach formula, see below).
5. Rinse in distilled water for 6 min (4 changes; to the last change, add 2 ml Kodak Photo-Flo per liter).
6. Dry at 20°C and 40–60% relative humidity.

Developer formula

metol	2 g
sodium sulfite (anhydrous)	95 g
hydroquinone	8 g
sodium carbonate (monohydrate)	45 g
potassium bromide	5 g
sodium thiosulfate	0.5 g
distilled water to make	1 l.

Bleach formula

Stock solution A	
potassium dichromate	8 g
conc. sulfuric acid	10 ml
distilled water to make	1 l.
Stock solution B	
potassium iodide	2 g
distilled water to make	1 l.

Mix 1 part of A, 1 part of B and 8 parts of distilled water just before use. All solutions at 20 ± 0.5°C.

in scattering is possible through controlled etching of the grains in a dilute solution of a solvent, such as sodium thiosulfate, to reduce their size.¹⁷ This can be done without an additional processing step if this etch bath is combined with the developer.¹⁸ Under these conditions, some of the unexposed silver halide dissolved by the thiosulfate is reduced by the developer and a certain amount of physical development takes place at adjacent development centers. This speeds up the rate of attack of the unexposed grains by the thiosulfate in the vicinity of the developing grains, resulting in enhanced modulation in the image. There is, of course, an increase in the amount of image silver formed, but this is not important, since it is eliminated by the bleach.

With this technique it is possible to produce holograms in which the ratio of the intensity of the image of a diffusely reflecting object to that of the nominally black background is better than 60:1. This compares very favorably with values obtained with unbleached holograms.

Another major problem with bleached holograms has been their tendency to darken when exposed to ambient light because of the formation of printout silver. Their resistance to exposure to light can be improved by treatment with a desensitizing dye¹⁵ but this has the drawback that it stains the gelatin. It has been shown, however, that the silver halide of which the image is formed also determines its stability against printout, AgCl being the poorest in this respect, while AgBr is considerably better and AgI exhibits a very high degree of resistance.¹⁹ It is possible to convert an image consisting of AgCl or AgBr to AgI, with very good results, by soaking it in a dilute solution of KI,²⁰ though this involves an extra step in processing. The results of comparative measurements of stability against exposure to light of untreated and treated holograms are presented in Fig. 4.

Rather surprisingly, it was found possible to combine this step with the previous one. Very bright images were obtained

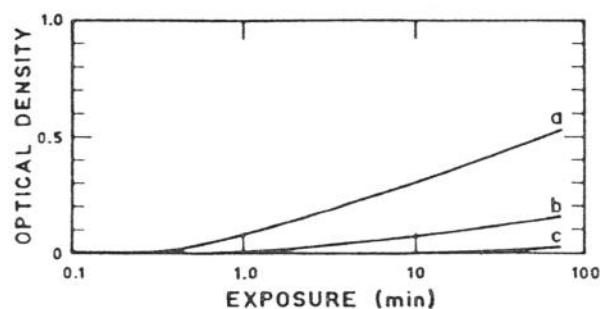


Figure 4. Increase in optical density of bleached holograms when exposed to direct noon sunlight. (a) Bleach containing KBr; (b) bleach followed by treatment in 0.2% KI; (c) bleach containing 0.2% KI.

with a dichromate bleach containing a small amount of KI, though from first considerations one would expect this treatment to result in a very weak phase image.²¹

An analysis suggests that several processes are taking place side by side in this bath. The first is oxidation of the silver image by the bleach to Ag⁺. The second is dissolution of the unexposed silver halide grains, due to the finite though low solubility of silver halides in KI solutions, accompanied by progressive conversion of the unexposed silver halide grains to AgI. If the concentration of KI in the bath is adjusted so that the solubility of AgI is in the proper range, the solution is rapidly saturated with respect to AgI. The build up of Ag⁺ is then followed by deposition of Ag⁺ and I⁻ on the nuclei provided by the unexposed silver halide grains, resulting in the formation of an intensified phase image.

The final result is an extremely simple processing technique for bleached phase holograms involving only two baths. Table II shows a typical processing schedule for Scientia 8E75 plates. Plates processed according to this have consistently given holograms with good diffraction efficiency, very low scattering and very good stability against exposure to light.

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